

Econometric Supplement for “Estimation and inference in discrete games with uncertain behavior”

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Abstract

This document includes step-by-step derivations leading to the main econometric results in the paper. Every section in this document has the format **SX** and every equation has the format **(SX.X)**. Any section, assumption or equation that we reference here which does not have this format refers to the main paper, or to Appendix A (if it has the format A.X).

S1 Asymptotic properties of $\widehat{\Delta}$ and proof of Proposition 1

Recall that,

$$\begin{aligned}\mu_S^C(X, \theta) &\equiv F_1(X_1^{ns'} \beta_1) + F_2(X_2^{ns'} \beta_2) - F_{1,2}(X_1^{ns'} \beta_1, X_2^{ns'} \beta_2 | \rho) \\ &\quad + F_{1,2}(X_1^{ns'} \beta_1 - X_1^{s'} \Delta_1 - X_2^{s'} \Delta_2, X_2^{ns'} \beta_2 - X_1^{s'} \Delta_1 - X_2^{s'} \Delta_2 | \rho), \\ \mu_S^{NC}(X, \theta) &\equiv F_1(X_1^{ns'} \beta_1) + F_2(X_2^{ns'} \beta_2) - F_{1,2}(X_1^{ns'} \beta_1, X_2^{ns'} \beta_2 | \rho) \\ &\quad + F_{1,2}(X_1^{ns'} \beta_1 - X_1^{s'} \Delta_1, X_2^{ns'} \beta_2 - X_2^{s'} \Delta_2 | \rho).\end{aligned}$$

and $E[S|U] = \mu_S^{NC}(X, \theta_0) + \pi(Z) \cdot \Xi_S(X, \theta_0)$, where $\Xi_S(X, \theta) \equiv \mu_S^C(X, \theta) - \mu_S^{NC}(X, \theta)$. We begin by describing the asymptotic properties of the semiparametric convolution weights $\widehat{\pi}(z, \theta)$ introduced in equation (14) of the paper.

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S1.1 Asymptotic properties of $\widehat{\pi}(z, \theta)$

Take the functionals we defined in Assumption E2.

$$\begin{aligned}
\mu_I^{\Xi_S}(z, \theta) &\equiv E[(S - \mu_S^{NC}(X, \theta)) \cdot \Xi_S(X, \theta) | Z = z] \\
\mu_{II, \kappa}^{\Xi_S}(z, \theta) &\equiv E[\Xi_S(X, \theta)^\kappa | Z = z] \text{ (for } \kappa = 1, 2) \\
\mu_{III, \theta_\ell}^{\Xi_S}(z, \theta) &\equiv E\left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot (S - \mu_S^{NC}(X, \theta)) \cdot \middle| Z = z\right] \text{ (for } \ell = 1, \dots, d_\theta) \\
\mu_{IV, \theta_\ell}^{\Xi_S}(z, \theta) &\equiv E\left[\frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X, \theta) \middle| Z = z\right] \text{ (for } \ell = 1, \dots, d_\theta) \\
\mu_{V, \theta_\ell}^{\Xi_S}(z, \theta) &\equiv E\left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X, \theta) \middle| Z = z\right] \text{ (for } \ell = 1, \dots, d_\theta) \\
\mu_{VI, \theta_\ell}^{\Xi_S}(z, \theta) &\equiv E\left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \middle| Z = z\right] \text{ (for } \ell = 1, \dots, d_\theta)
\end{aligned} \tag{S1.1}$$

From here, for any $z \in \text{Supp}(Z)$ (i.e, any z such that $f_Z(z) > 0$) and any $\theta \in \Theta$ we will denote

$$R_a^\pi(z, \theta) \equiv \mu_I^{\Xi_S}(z, \theta) \cdot f_Z(z) \quad \text{and} \quad R_b^\pi(z, \theta) \equiv \mu_{II, 2}^{\Xi_S}(z, \theta) \cdot f_Z(z).$$

For any $z \in \text{Supp}(Z)$, we can write

$$\pi(z, \theta) = \frac{R_a^\pi(z, \theta)}{R_b^\pi(z, \theta)}. \tag{S1.2}$$

We can describe our estimator $\widehat{\pi}(z, \theta)$ using sample analogs of $R_a^\pi(z, \theta)$ and $R_b^\pi(z, \theta)$. Let

$$\begin{aligned}
\widehat{R}_a^\pi(z, \theta) &\equiv \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n (S_i - \mu_S^{NC}(X_i, \theta)) \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right), \\
\widehat{R}_b^\pi(z, \theta) &\equiv \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \Xi_S(X_i, \theta)^2 \cdot K\left(\frac{Z_i - z}{h_n}\right).
\end{aligned}$$

Take a given (z, θ) . As described in equation (14) of the paper, let

$$\widehat{\pi}(z, \theta) \equiv \frac{\widehat{R}_a^\pi(z, \theta)}{\widehat{R}_b^\pi(z, \theta)}. \tag{S1.3}$$

Next, for a given $h > 0$ let,

$$\begin{aligned} v_{a,n}^\pi(z, \theta; h) &\equiv \frac{1}{n} \sum_{i=1}^n \left((S_i - \mu_S^{NC}(X_i, \theta)) \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h}\right) - E\left[(S - \mu_S^{NC}(X, \theta)) \cdot \Xi_S(X, \theta) \cdot K\left(\frac{Z - z}{h}\right) \right] \right), \\ v_{b,n}^\pi(z, \theta; h) &\equiv \frac{1}{n} \sum_{i=1}^n \left(\Xi_S(X_i, \theta)^2 \cdot K\left(\frac{Z_i - z}{h}\right) - E\left[\Xi_S(X, \theta)^2 \cdot K\left(\frac{Z - z}{h}\right) \right] \right) \end{aligned}$$

By the bounded-variation properties of our kernel function described in Assumption E1 and the Lipschitz-restrictions in Assumption E3, Nolan and Pollard (1987, Lemma 22), Pakes and Pollard (1989, Example 2.10, Lemma 2.13, Lemma 2.14) imply that the class of functions involved is *Euclidean* (see Sherman (1994, Definition 3)),

$$\begin{aligned} \mathcal{G}_a &\equiv \left\{ g(s, x, z) = (s - \mu_S^{NC}(x, \theta)) \cdot \Xi_S(x, \theta) \cdot K\left(\frac{z - \tilde{z}}{h}\right) \text{ for some } \theta \in \Theta, \tilde{z} \in \mathbb{R}^{d_z}, h > 0 \right\}, \\ \mathcal{G}_b &\equiv \left\{ g(x, z) = \Xi_S(x, \theta)^2 \cdot K\left(\frac{z - \tilde{z}}{h}\right) \text{ for some } \theta \in \Theta, \tilde{z} \in \mathbb{R}^{d_z}, h > 0 \right\}. \end{aligned}$$

Since $|\Xi_S(x, \theta)| \leq 1$ and $|\mu_S^{NC}(x, \theta)| \leq 1$ for all (x, θ) , $0 \leq S \leq 2$, and $|K(\psi)| \leq \bar{K}$ for all ψ , the corresponding envelope for the class of functions is constant, and thus trivially has finite $2 + \delta$ moments. From here, Sherman (1994, Corollary 4) yields,

$$\sup_{\substack{\theta \in \Theta \\ z \in \mathbb{R}^{d_z} \\ h > 0}} |v_{a,n}^\pi(z, \theta; h)| = O_p(n^{-1/2}), \quad \sup_{\substack{\theta \in \Theta \\ z \in \mathbb{R}^{d_z} \\ h > 0}} |v_{b,n}^\pi(z, \theta; h)| = O_p(n^{-1/2})$$

Next, note that we can express,

$$\begin{aligned} \widehat{R}_a^\pi(z, \theta) &= R_a^\pi(z, \theta) + B_{a,n}^\pi(z, \theta) + \frac{1}{h_n^{d_z}} \cdot v_{a,n}^\pi(z, \theta; h_n), \\ \widehat{R}_b^\pi(z, \theta) &= R_b^\pi(z, \theta) + B_{b,n}^\pi(z, \theta) + \frac{1}{h_n^{d_z}} \cdot v_{b,n}^\pi(z, \theta; h_n), \end{aligned} \tag{S1.4}$$

where $B_{a,n}^\pi(z, \theta)$ and $B_{b,n}^\pi(z, \theta)$ represent the bias terms,

$$\begin{aligned} B_{a,n}^\pi(z, \theta) &\equiv E\left[\frac{1}{h_n^{d_z}} \cdot (S - \mu_S^{NC}(X, \theta)) \cdot \Xi_S(X, \theta) \cdot K\left(\frac{Z - z}{h_n}\right) \right] - R_a^\pi(z, \theta), \\ B_{b,n}^\pi(z, \theta) &\equiv E\left[\frac{1}{h_n^{d_z}} \cdot \Xi_S(X, \theta)^2 \cdot K\left(\frac{Z - z}{h_n}\right) \right] - R_b^\pi(z, \theta). \end{aligned}$$

An M^{th} -order approximation combined with the smoothness restrictions in Assumption

E2 and the higher-order properties of the kernel described in Assumption E1 yield,

$$\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |B_{a,n}^\pi(z, \theta)| = O(h_n^M), \quad \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |B_{b,n}^\pi(z, \theta)| = O(h_n^M)$$

Combined with our bandwidth convergence restrictions in Assumption E1, these results imply

$$\begin{aligned} \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\widehat{R}_a^\pi(z, \theta) - R_a^\pi(z, \theta)| &= O(h_n^M) + O_p\left(\frac{1}{h_n^{d_Z} \cdot n^{1/2}}\right) = o_p(n^{-1/4}), \\ \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\widehat{R}_b^\pi(z, \theta) - R_b^\pi(z, \theta)| &= O(h_n^M) + O_p\left(\frac{1}{h_n^{d_Z} \cdot n^{1/2}}\right) = o_p(n^{-1/4}), \end{aligned} \quad (\text{S1.5})$$

Recall that we denote $U \equiv X \cup Z$ and $V \equiv (S, U)$. For a given (z, θ) , define

$$\begin{aligned} \psi_{a,n}^\pi(V_i; z, \theta) &\equiv (S_i - \mu_S^{NC}(X_i, \theta)) \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right) - E\left[(S - \mu_S^{NC}(X, \theta)) \cdot \Xi_S(X, \theta) \cdot K\left(\frac{Z - z}{h_n}\right)\right], \\ \psi_{b,n}^\pi(U_i; z, \theta) &\equiv \Xi_S(X_i, \theta)^2 \cdot K\left(\frac{Z_i - z}{h_n}\right) - E\left[\Xi_S(X, \theta)^2 \cdot K\left(\frac{Z - z}{h_n}\right)\right] \end{aligned} \quad (\text{S1.6})$$

Note that $E[\psi_{a,n}^\pi(V_i; z, \theta)] = 0$ and $E[\psi_{b,n}^\pi(U_i; z, \theta)] = 0 \forall (z, \theta)$. The expressions in (S1.4) can be expressed as linear representation results for $\widehat{R}_a^\pi(z, \theta)$ and $\widehat{R}_b^\pi(z, \theta)$, given by

$$\begin{aligned} \widehat{R}_a^\pi(z, \theta) &= R_a^\pi(z, \theta) + \frac{1}{n \cdot h_n^{d_Z}} \cdot \sum_{i=1}^n \psi_{a,n}^\pi(V_i; z, \theta) + B_{a,n}^\pi(z, \theta), \quad \text{where} \quad \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |B_{a,n}^\pi(z, \theta)| = O(h_n^M), \\ \widehat{R}_b^\pi(z, \theta) &= R_b^\pi(z, \theta) + \frac{1}{n \cdot h_n^{d_Z}} \cdot \sum_{i=1}^n \psi_{b,n}^\pi(U_i; z, \theta) + B_{b,n}^\pi(z, \theta), \quad \text{where} \quad \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |B_{b,n}^\pi(z, \theta)| = O(h_n^M) \end{aligned} \quad (\text{S1.7})$$

Note from our bandwidth convergence restrictions in Assumption E1 that,

$$\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |B_{a,n}^\pi(z, \theta)| = O(h_n^M) = o(n^{-1/2}), \quad \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |B_{b,n}^\pi(z, \theta)| = O(h_n^M) = o(n^{-1/2}). \quad (\text{S1.8})$$

Recall from Assumption E2 that there exists a $C > 0$ such that $R_b^\pi(z, \theta) \geq C^2$ for all $(z, \theta) \in \mathcal{Z} \times \Theta$. Using this restriction, equations (S1.2) and (S1.3) and a second-order approximation yields

$$\begin{aligned} \widehat{\pi}(z, \theta) &= \pi(z, \theta) + \frac{1}{R_b^\pi(z, \theta)} \cdot (\widehat{R}_a^\pi(z, \theta) - R_a^\pi(z, \theta)) - \frac{\pi(z, \theta)}{R_b^\pi(z, \theta)} \cdot (\widehat{R}_b^\pi(z, \theta) - R_b^\pi(z, \theta)) \\ &\quad + O_p\left(\left\|(\widehat{R}_a^\pi(z, \theta) - R_a^\pi(z, \theta), \widehat{R}_b^\pi(z, \theta) - R_b^\pi(z, \theta))\right\|^2\right) \end{aligned} \quad (\text{S1.9})$$

From (S1.5), we have

$$\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left\| \left(\widehat{R}_a^\pi(z, \theta) - R_a^\pi(z, \theta), \widehat{R}_b^\pi(z, \theta) - R_b^\pi(z, \theta) \right) \right\|^2 = O_p \left(\frac{1}{(h_n^{d_Z} \cdot n^{1/2})^2} \right).$$

Using this result and plugging in the linear representations in (S1.7), the expression in (S1.9) yields,

$$\begin{aligned} \widehat{\pi}(z, \theta) &= \pi(z, \theta) + \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \psi_n^\pi(V_i; z, \theta) + \vartheta_n^\pi(z, \theta), \quad \text{where} \\ \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\vartheta_n^\pi(z, \theta)| &= O(h_n^M) + O_p \left(\frac{1}{(h_n^{d_Z} \cdot n^{1/2})^2} \right) = o_p(n^{-1/2}), \quad \text{and} \\ \psi_n^\pi(V_i; z, \theta) &\equiv \frac{1}{R_b^\pi(z, \theta)} \cdot \psi_{a,n}^\pi(V_i; z, \theta) - \frac{\pi(z, \theta)}{R_b^\pi(z, \theta)} \cdot \psi_{b,n}^\pi(U_i; z, \theta) \end{aligned} \quad (\text{S1.10})$$

where $\psi_{a,n}^\pi(V_i; z, \theta)$ and $\psi_{b,n}^\pi(U_i; z, \theta)$ are the influence functions described in (S1.6). The result $O(h_n^M) + O_p \left(\frac{1}{(h_n^{d_Z} \cdot n^{1/2})^2} \right) = o_p(n^{-1/2})$ follows from the bandwidth convergence restrictions in Assumption E1. Equation (S1.10) is a linear representation result for $\widehat{\pi}(z, \theta)$ and will be invoked repeatedly below. Note that, since $E[\psi_{a,n}^\pi(V_i; z, \theta)] = 0$ and $E[\psi_{b,n}^\pi(U_i; z, \theta)] = 0 \forall (z, \theta)$, we have $E[\psi_n^\pi(V_i; z, \theta)] = 0$ for all (z, θ) such that $R_b^\pi(z, \theta) \neq 0$. In particular, we have $E[\psi_n^\pi(V_i; z, \theta)] = 0 \forall (z, \theta) \in \mathcal{Z} \times \Theta$. From (S1.10) it also follows that,

$$\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\widehat{\pi}(z, \theta) - \pi(z, \theta)| = O_p \left(\frac{1}{h_n^{d_Z} \cdot n^{1/2}} \right) + o_p(n^{-1/2}) = o_p(n^{-1/4}). \quad (\text{S1.11})$$

where the last result follows from the bandwidth convergence restrictions in Assumption E1. Next, we study the properties of $\frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_\ell} - \frac{\partial \pi(z, \theta)}{\partial \theta_\ell}$ for $\ell = 1, \dots, d_\theta$. From the expressions in (S1.2) and (S1.3), we have

$$\begin{aligned} \frac{\partial \pi(z, \theta)}{\partial \theta_\ell} &= \frac{1}{R_b^\pi(z, \theta)} \cdot \left(\frac{\partial R_a^\pi(z, \theta)}{\partial \theta_\ell} - \pi(z, \theta) \cdot \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \right) \\ \frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_\ell} &= \frac{1}{\widehat{R}_b^\pi(z, \theta)} \cdot \left(\frac{\partial \widehat{R}_a^\pi(z, \theta)}{\partial \theta_\ell} - \widehat{\pi}(z, \theta) \cdot \frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} \right) \end{aligned} \quad (\text{S1.12})$$

Note that, using the functional definitions in (S1.1),

$$\begin{aligned}
\frac{\partial R_a^\pi(z, \theta)}{\partial \theta_\ell} &= E \left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot (S - \mu_S^{NC}(X, \theta)) \mid Z = z \right] \cdot f_Z(z) - E \left[\frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X, \theta) \mid Z = z \right] \cdot f_Z(z) \\
&\equiv (\mu_{III, \theta_\ell}^{\Xi_S}(z, \theta) - \mu_{IV, \theta_\ell}^{\Xi_S}(z, \theta)) \cdot f_Z(z) \\
\frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} &= 2 \cdot E \left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X, \theta) \mid Z = z \right] \cdot f_Z(z) \\
&\equiv 2 \cdot \mu_{V, \theta_\ell}^{\Xi_S}(z, \theta) \cdot f_Z(z).
\end{aligned}$$

We have already characterized the asymptotic properties of $\widehat{R}_b^\pi(z, \theta)$ and $\widehat{\pi}(z, \theta)$. We now proceed to study the asymptotic properties of $\frac{\partial \widehat{R}_a^\pi(z, \theta)}{\partial \theta_\ell}$ and $\frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell}$. We have

$$\begin{aligned}
\frac{\partial \widehat{R}_a^\pi(z, \theta)}{\partial \theta_\ell} &= \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \left(\frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot (S_i - \mu_S^{NC}(X_i, \theta)) - \frac{\partial \mu_S^{NC}(X_i, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X_i, \theta) \right) \cdot K \left(\frac{Z_i - z}{h_n} \right) \\
\frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} &= \frac{2}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X_i, \theta) \cdot K \left(\frac{Z_i - z}{h_n} \right)
\end{aligned}$$

Let

$$\begin{aligned}
\psi_{a,n}^{\nabla \theta_\ell \pi}(V_i; z, \theta) &\equiv \left(\frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot (S_i - \mu_S^{NC}(X_i, \theta)) - \frac{\partial \mu_S^{NC}(X_i, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X_i, \theta) \right) \cdot K \left(\frac{Z_i - z}{h_n} \right) \\
&\quad - E \left[\left(\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot (S - \mu_S^{NC}(X, \theta)) - \frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X, \theta) \right) \cdot K \left(\frac{Z - z}{h_n} \right) \right] \\
\psi_{b,n}^{\nabla \theta_\ell \pi}(U_i; z, \theta) &\equiv 2 \cdot \left(\frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X_i, \theta) \cdot K \left(\frac{Z_i - z}{h_n} \right) - E \left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \Xi_S(X, \theta) \cdot K \left(\frac{Z - z}{h_n} \right) \right] \right), \\
&\hspace{15em} (S1.13)
\end{aligned}$$

Note that $E[\psi_{a,n}^{\nabla \theta_\ell \pi}(V; z, \theta)] = 0$ and $E[\psi_{b,n}^{\nabla \theta_\ell \pi}(U; z, \theta)] = 0$ for any (z, θ) . Next, define the bias

terms,

$$\begin{aligned}
B_{a,n}^{\nabla\theta_\ell\pi}(z,\theta) &\equiv \\
E \left[\frac{1}{h_n^{d_Z}} \cdot \left(\frac{\partial \Xi_S(X,\theta)}{\partial \theta_\ell} \cdot (S - \mu_S^{NC}(X,\theta)) - \frac{\partial \mu_S^{NC}(X,\theta)}{\partial \theta_\ell} \cdot \Xi_S(X,\theta) \right) \cdot K \left(\frac{Z-z}{h_n} \right) \right] &- (\mu_{III,\theta_\ell}^{\Xi_S}(z,\theta) - \mu_{IV,\theta_\ell}^{\Xi_S}(z,\theta)) \cdot f_Z(z) \\
= E \left[\frac{1}{h_n^{d_Z}} \cdot \left(\frac{\partial \Xi_S(X,\theta)}{\partial \theta_\ell} \cdot (S - \mu_S^{NC}(X,\theta)) - \frac{\partial \mu_S^{NC}(X,\theta)}{\partial \theta_\ell} \cdot \Xi_S(X,\theta) \right) \cdot K \left(\frac{Z-z}{h_n} \right) \right] &- \frac{\partial R_a^\pi(z,\theta)}{\partial \theta_\ell},
\end{aligned}$$

$$\begin{aligned}
B_{b,n}^{\nabla\theta_\ell\pi}(z,\theta) &\equiv \\
2 \cdot E \left[\frac{1}{h_n^{d_Z}} \cdot \frac{\partial \Xi_S(X,\theta)}{\partial \theta_\ell} \cdot \Xi_S(X,\theta) \cdot K \left(\frac{Z-z}{h_n} \right) \right] &- 2 \cdot \mu_{V,\theta_\ell}^{\Xi_S}(z,\theta) \cdot f_Z(z) \\
= 2 \cdot E \left[\frac{1}{h_n^{d_Z}} \cdot \frac{\partial \Xi_S(X,\theta)}{\partial \theta_\ell} \cdot \Xi_S(X,\theta) \cdot K \left(\frac{Z-z}{h_n} \right) \right] &- \frac{\partial R_b^\pi(z,\theta)}{\partial \theta_\ell}
\end{aligned}$$

An M^{th} -order approximation combined with the smoothness restrictions in Assumption E2 and the higher-order properties of the kernel described in Assumption E1 yield,

$$\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |B_{a,n}^{\nabla\theta_\ell\pi}(z,\theta)| = O(h_n^M) = o(n^{-1/2}), \quad \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |B_{b,n}^{\nabla\theta_\ell\pi}(z,\theta)| = O(h_n^M) = o(n^{-1/2}), \quad (S1.14)$$

where (as before), the result $O(h_n^M) = o(n^{-1/2})$ follows from the bandwidth convergence restrictions in Assumption E1. From here, repeating the arguments that led to the linear representations in (S1.7), our restrictions now yield the following linear representations for $\frac{\partial \widehat{R}_a^\pi(z,\theta)}{\partial \theta_\ell}$ and $\frac{\partial \widehat{R}_b^\pi(z,\theta)}{\partial \theta_\ell}$,

$$\begin{aligned}
\frac{\partial \widehat{R}_a^\pi(z,\theta)}{\partial \theta_\ell} &= \frac{\partial R_a^\pi(z,\theta)}{\partial \theta_\ell} + \frac{1}{n \cdot h_n^{d_Z}} \cdot \sum_{i=1}^n \psi_{a,n}^{\nabla\theta_\ell\pi}(V_i; z, \theta) + B_{a,n}^{\nabla\theta_\ell\pi}(z, \theta), \quad \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |B_{a,n}^{\nabla\theta_\ell\pi}(z, \theta)| = o_p(n^{-1/2}), \\
\frac{\partial \widehat{R}_b^\pi(z,\theta)}{\partial \theta_\ell} &= \frac{\partial R_b^\pi(z,\theta)}{\partial \theta_\ell} + \frac{1}{n \cdot h_n^{d_Z}} \cdot \sum_{i=1}^n \psi_{b,n}^{\nabla\theta_\ell\pi}(U_i; z, \theta) + B_{b,n}^{\nabla\theta_\ell\pi}(z, \theta), \quad \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |B_{b,n}^{\nabla\theta_\ell\pi}(z, \theta)| = o_p(n^{-1/2})
\end{aligned} \quad (S1.15)$$

From the above result, we also have

$$\begin{aligned}
\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial \widehat{R}_a^\pi(z,\theta)}{\partial \theta_\ell} - \frac{\partial R_a^\pi(z,\theta)}{\partial \theta_\ell} \right| &= O_p \left(\frac{1}{h_n^{d_Z} \cdot n^{1/2}} \right) + o_p(n^{-1/2}) = o_p(n^{-1/4}), \\
\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial \widehat{R}_b^\pi(z,\theta)}{\partial \theta_\ell} - \frac{\partial R_b^\pi(z,\theta)}{\partial \theta_\ell} \right| &= O_p \left(\frac{1}{h_n^{d_Z} \cdot n^{1/2}} \right) + o_p(n^{-1/2}) = o_p(n^{-1/4})
\end{aligned} \quad (S1.16)$$

We are now ready to describe a linear representation result for $\frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_\ell} - \frac{\partial \pi(z, \theta)}{\partial \theta_\ell}$. To this end, we go back (S1.12). Note first that,

$$\begin{aligned} \frac{\partial \widehat{R}_a^\pi(z, \theta)}{\partial \theta_\ell} - \widehat{\pi}(z, \theta) \cdot \frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} &= \frac{R_a^\pi(z, \theta)}{\partial \theta_\ell} - \pi(z, \theta) \cdot \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} + \left(\frac{\partial \widehat{R}_a^\pi(z, \theta)}{\partial \theta_\ell} - \frac{\partial R_a^\pi(z, \theta)}{\partial \theta_\ell} \right) \\ &\quad - \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \cdot (\widehat{\pi}(z, \theta) - \pi(z, \theta)) - \pi(z, \theta) \cdot \left(\frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} - \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \right) - \xi_{I,n}^{\nabla \theta_\ell \pi}(z, \theta), \end{aligned}$$

where $\xi_{I,n}^{\nabla \theta_\ell \pi}(z, \theta) \equiv -(\widehat{\pi}(z, \theta) - \pi(z, \theta)) \cdot \left(\frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} - \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \right)$

From (S1.11) and (S1.16),

$$\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \xi_{I,n}^{\nabla \theta_\ell \pi}(z, \theta) \right| = o_p(n^{-1/2}). \quad (\text{S1.17})$$

Next, we have $\frac{1}{\widehat{R}_b^\pi(z, \theta)} = \frac{1}{R_b^\pi(z, \theta)} - \frac{1}{R_b^\pi(z, \theta)^2} \cdot (\widehat{R}_b^\pi(z, \theta) - R_b^\pi(z, \theta)) + \xi_{II,n}^{\nabla \theta_\ell \pi}(z, \theta)$, where $\xi_{II,n}^{\nabla \theta_\ell \pi}(z, \theta) = O_p\left(\left|\widehat{R}_b^\pi(z, \theta) - R_b^\pi(z, \theta)\right|^2\right) \forall (z, \theta) \in \mathcal{Z} \times \Theta$. From (S1.5),

$$\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \xi_{II,n}^{\nabla \theta_\ell \pi}(z, \theta) \right| = o_p(n^{-1/2}). \quad (\text{S1.18})$$

From here, going back to (S1.12), we obtain,

$$\begin{aligned} \frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_\ell} &= \frac{\partial \pi(z, \theta)}{\partial \theta_\ell} + \frac{1}{R_b^\pi(z, \theta)} \times \left\{ \left(\frac{\partial \widehat{R}_a^\pi(z, \theta)}{\partial \theta_\ell} - \frac{\partial R_a^\pi(z, \theta)}{\partial \theta_\ell} \right) - \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \cdot (\widehat{\pi}(z, \theta) - \pi(z, \theta)) \right. \\ &\quad \left. - \pi(z, \theta) \cdot \left(\frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} - \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \right) - \frac{1}{R_b^\pi(z, \theta)} \cdot \left(\frac{\partial R_a^\pi(z, \theta)}{\partial \theta_\ell} - \pi(z, \theta) \cdot \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \right) \cdot (\widehat{R}_b^\pi(z, \theta) - R_b^\pi(z, \theta)) \right\} \\ &\quad + \xi_{III,n}^{\nabla \theta_\ell \pi}(z, \theta), \quad \text{where} \\ \xi_{III,n}^{\nabla \theta_\ell \pi}(z, \theta) &\equiv \left\{ -\frac{1}{R_b^\pi(z, \theta)^2} \cdot (\widehat{R}_b^\pi(z, \theta) - R_b^\pi(z, \theta)) + \xi_{II,n}^{\nabla \theta_\ell \pi}(z, \theta) \right\} \\ &\quad \times \left\{ \left(\frac{\partial \widehat{R}_a^\pi(z, \theta)}{\partial \theta_\ell} - \frac{\partial R_a^\pi(z, \theta)}{\partial \theta_\ell} \right) - \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \cdot (\widehat{\pi}(z, \theta) - \pi(z, \theta)) \right. \\ &\quad \left. - \pi(z, \theta) \cdot \left(\frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} - \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \right) - \frac{1}{R_b^\pi(z, \theta)} \cdot \left(\frac{\partial R_a^\pi(z, \theta)}{\partial \theta_\ell} - \pi(z, \theta) \cdot \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \right) \cdot (\widehat{R}_b^\pi(z, \theta) - R_b^\pi(z, \theta)) \right\} \\ &\quad + \xi_{II,n}^{\nabla \theta_\ell \pi}(z, \theta) \cdot \left(\frac{\partial R_a^\pi(z, \theta)}{\partial \theta_\ell} - \pi(z, \theta) \cdot \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \right) \end{aligned} \quad (\text{S1.19})$$

Using the results in equations (S1.5), (S1.11), (S1.16), (S1.17) and (S1.18), the boundedness

properties of the various functionals of Z described in Assumption E2 yield,

$$\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \xi_{III,n}^{\nabla \theta_\ell \pi}(z, \theta) \right| = o_p(n^{-1/2}). \quad (\text{S1.20})$$

The last step is to plug in the linear representation results in (S1.7), (S1.10) and (S1.15) into (S1.19). Let

$$\begin{aligned} \xi_{IV,n}^{\nabla \theta_\ell \pi}(z, \theta) \equiv & \frac{1}{R_b^\pi(z, \theta)} \times \left\{ B_{a,n}^{\nabla \theta_\ell \pi}(z, \theta) - \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \cdot \vartheta_n^\pi(z, \theta) - \pi(z, \theta) \cdot B_{b,n}^{\nabla \theta_\ell \pi}(z, \theta) \right. \\ & \left. - \frac{1}{R_b^\pi(z, \theta)} \cdot \left(\frac{\partial R_a^\pi(z, \theta)}{\partial \theta_\ell} - \pi(z, \theta) \cdot \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \right) \cdot B_{b,n}^\pi(z, \theta) \right\} \end{aligned}$$

where $B_{b,n}^\pi(z, \theta)$ has the properties stated in (S1.8), $B_{a,n}^{\nabla \theta_\ell \pi}(z, \theta)$ and $B_{b,n}^{\nabla \theta_\ell \pi}(z, \theta)$ have the properties stated in (S1.14) and $\vartheta_n^\pi(z, \theta)$ is as described in (S1.10). From those equations and the boundedness properties of the functionals described in Assumption E2, we obtain,

$$\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \xi_{IV,n}^{\nabla \theta_\ell \pi}(z, \theta) \right| = o_p(n^{-1/2}). \quad (\text{S1.21})$$

Let

$$\begin{aligned} \psi_n^{\nabla \theta_\ell \pi}(V_i; z, \theta) \equiv & \frac{1}{R_b^\pi(z, \theta)} \times \left\{ \psi_{a,n}^{\nabla \theta_\ell \pi}(V_i; z, \theta) - \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \cdot \psi_n^\pi(V_i; z, \theta) \right. \\ & \left. - \pi(z, \theta) \cdot \psi_{b,n}^{\nabla \theta_\ell \pi}(V_i; z, \theta) - \frac{1}{R_b^\pi(z, \theta)} \cdot \left(\frac{\partial R_a^\pi(z, \theta)}{\partial \theta_\ell} - \pi(z, \theta) \cdot \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \right) \cdot \psi_{b,n}^\pi(U_i; z, \theta) \right\}, \end{aligned} \quad (\text{S1.22})$$

where $\psi_{b,n}^\pi(U_i; z, \theta)$ is defined in (S1.6), $\psi_n^\pi(V_i; z, \theta)$ is defined in (S1.10), and $\psi_{a,n}^{\nabla \theta_\ell \pi}(V_i; z, \theta)$ and $\psi_{b,n}^{\nabla \theta_\ell \pi}(V_i; z, \theta)$ are as defined in (S1.13). Note from the properties of those functionals that $E[\psi_n^{\nabla \theta_\ell \pi}(V_i; z, \theta)] = 0 \forall (z, \theta) \in \mathcal{Z} \times \Theta$. Finally, let $\vartheta_n^{\nabla \theta_\ell \pi}(z, \theta) \equiv \xi_{III,n}^{\nabla \theta_\ell \pi}(z, \theta) + \xi_{IV,n}^{\nabla \theta_\ell \pi}(z, \theta)$ and note from (S1.20) and (S1.21) that,

$$\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \vartheta_n^{\nabla \theta_\ell \pi}(z, \theta) \right| = o_p(n^{-1/2}).$$

From (S1.19) and our previous linear representation results, we have,

$$\begin{aligned} \frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_\ell} &= \frac{\partial \pi(z, \theta)}{\partial \theta_\ell} + \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \psi_n^{\nabla \theta_\ell \pi}(V_i; z, \theta) + \mathfrak{S}_n^{\nabla \theta_\ell \pi}(z, \theta), \\ \text{where } \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \mathfrak{S}_n^{\nabla \theta_\ell \pi}(z, \theta) \right| &= o_p(n^{-1/2}). \end{aligned} \quad (\text{S1.23})$$

From (S1.23) it also follows that,

$$\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_\ell} - \frac{\partial \pi(z, \theta)}{\partial \theta_\ell} \right| = O_p\left(\frac{1}{h_n^{d_Z} \cdot n^{1/2}}\right) + o_p(n^{-1/2}) = o_p(n^{-1/4}). \quad (\text{S1.24})$$

Next let us study $\frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_j \partial \theta_\ell}$. To establish the asymptotic properties of our estimator, we only need to focus on the probability-limit of $\frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_j \partial \theta_\ell} - \frac{\partial \pi(z, \theta)}{\partial \theta_j \partial \theta_\ell}$ and not its asymptotic distribution. Take any $\ell, j = 1, \dots, d_\theta$. We have,

$$\begin{aligned} \frac{\partial^2 \pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} &= -\frac{1}{R_b^\pi(z, \theta)^2} \cdot \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_j} \cdot \left(\frac{\partial R_a^\pi(z, \theta)}{\partial \theta_\ell} - \pi(z, \theta) \cdot \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \right) \\ &\quad + \frac{1}{R_b^\pi(z, \theta)} \cdot \left(\frac{\partial^2 R_a^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} - \frac{\partial \pi(z, \theta)}{\partial \theta_j} \cdot \frac{R_b^\pi(z, \theta)}{\partial \theta_\ell} - \pi(z, \theta) \cdot \frac{\partial^2 R_b^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} \right) \end{aligned} \quad (\text{S1.25})$$

Take any $\ell, j = 1, \dots, d_\theta$. We have,

$$\begin{aligned} \frac{\partial^2 \widehat{\pi}(z, \theta)}{\partial \theta_j \partial \theta_\ell} &= -\frac{1}{\widehat{R}_b^\pi(z, \theta)^2} \cdot \frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_j} \cdot \left(\frac{\partial \widehat{R}_a^\pi(z, \theta)}{\partial \theta_\ell} - \widehat{\pi}(z, \theta) \cdot \frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} \right) \\ &\quad + \frac{1}{\widehat{R}_b^\pi(z, \theta)} \cdot \left(\frac{\partial^2 \widehat{R}_a^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} - \frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_j} \cdot \frac{\widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} - \widehat{\pi}(z, \theta) \cdot \frac{\partial^2 \widehat{R}_b^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} \right) \end{aligned} \quad (\text{S1.26})$$

From our previous results in equations (S1.5)-(S1.24), we have already established that,

$$\begin{aligned} \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{1}{\widehat{R}_b^\pi(z, \theta)^2} - \frac{1}{R_b^\pi(z, \theta)^2} \right| &= o_p(1), \quad \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial \widehat{R}_b^\pi(z, \theta)}{\partial \theta_\ell} - \frac{\partial R_b^\pi(z, \theta)}{\partial \theta_\ell} \right| = o_p(1) \quad (\text{for } \ell = 1, \dots, d_\theta) \\ \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial \widehat{R}_a^\pi(z, \theta)}{\partial \theta_\ell} - \frac{\partial R_a^\pi(z, \theta)}{\partial \theta_\ell} \right| &= o_p(1) \quad (\text{for } \ell = 1, \dots, d_\theta), \quad \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\widehat{\pi}(z, \theta) - \pi(z, \theta)| = o_p(1), \\ \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_\ell} - \frac{\partial \pi(z, \theta)}{\partial \theta_\ell} \right| &= o_p(1) \quad (\text{for } \ell = 1, \dots, d_\theta). \end{aligned} \quad (\text{S1.27})$$

Comparing (S1.25) and (S1.26), to obtain $\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial^2 \widehat{\pi}(z, \theta)}{\partial \theta_j \partial \theta_\ell} - \frac{\partial^2 \pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} \right| = o_p(1)$ for $j, \ell = 1, \dots, d_\theta$,

we only need to show $\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial^2 \widehat{R}_a^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} - \frac{\partial^2 R_a^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} \right| = o_p(1)$ and $\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial^2 \widehat{R}_b^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} - \frac{\partial^2 R_b^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} \right| = o_p(1)$.

Take the functionals we defined in Assumption E2.

$$\begin{aligned} \delta_{I, \theta_\ell, \theta_j}(z, \theta) &\equiv E \left[\frac{\partial^2 \Xi_S(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot (S - \mu_S^{NC}(X, \theta)) \mid Z = z \right] \text{ (for } j, \ell = 1, \dots, d_\theta) \\ \delta_{II, \theta_\ell, \theta_j}(z, \theta) &\equiv E \left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_j} \mid Z = z \right] \text{ (for } j, \ell = 1, \dots, d_\theta) \\ \delta_{III, \theta_\ell, \theta_j}(z, \theta) &\equiv E \left[\frac{\partial^2 \mu_S^{NC}(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X, \theta) \mid Z = z \right] \text{ (for } j, \ell = 1, \dots, d_\theta) \\ \delta_{IV, \theta_\ell, \theta_j}(z, \theta) &\equiv E \left[\frac{\partial^2 \Xi_S(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X, \theta) \mid Z = z \right] \text{ (for } j, \ell = 1, \dots, d_\theta) \\ \delta_{V, \theta_\ell, \theta_j}(z, \theta) &\equiv E \left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_j} \mid Z = z \right] \text{ (for } j, \ell = 1, \dots, d_\theta) \end{aligned}$$

We have,

$$\begin{aligned} \frac{\partial^2 R_a^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} &= (\delta_{I, \theta_\ell, \theta_j}(z, \theta) - \delta_{II, \theta_\ell, \theta_j}(z, \theta) - \delta_{III, \theta_\ell, \theta_j}(z, \theta) - \delta_{IV, \theta_\ell, \theta_j}(z, \theta)) \cdot f_Z(z), \\ \frac{\partial^2 R_b^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} &= 2 \cdot (\delta_{IV, \theta_\ell, \theta_j}(z, \theta) + \delta_{V, \theta_\ell, \theta_j}(z, \theta)) \cdot f_Z(z). \end{aligned}$$

And,

$$\begin{aligned} \frac{\partial^2 \widehat{R}_a^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} &= \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \frac{\partial^2 \Xi_S(X_i, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot (S_i - \mu_S^{NC}(X_i, \theta)) \cdot K\left(\frac{Z_i - z}{h_n}\right) \\ &\quad - \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \mu_S^{NC}(X_i, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z_i - z}{h_n}\right) \\ &\quad - \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \frac{\partial^2 \mu_S^{NC}(X_i, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right) \\ &\quad - \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \frac{\partial \mu_S^{NC}(X_i, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z_i - z}{h_n}\right), \\ \frac{\partial^2 \widehat{R}_b^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} &= \frac{2}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \frac{\partial^2 \Xi_S(X_i, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right) \\ &\quad + \frac{2}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z_i - z}{h_n}\right). \end{aligned} \tag{S1.28}$$

From here, we can express

$$\begin{aligned}
\frac{\partial^2 \widehat{R}_a^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} &= \frac{\partial^2 R_a^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} \\
&+ \frac{1}{n \cdot h_n^{dz}} \sum_{i=1}^n \left(\frac{\partial^2 \Xi_S(X_i, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot (S_i - \mu_S^{NC}(X_i, \theta)) \cdot K\left(\frac{Z_i - z}{h_n}\right) - E \left[\frac{\partial^2 \Xi_S(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot (S - \mu_S^{NC}(X, \theta)) \cdot K\left(\frac{Z - z}{h_n}\right) \right] \right) \\
&- \frac{1}{n \cdot h_n^{dz}} \sum_{i=1}^n \left(\frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \mu_S^{NC}(X_i, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z_i - z}{h_n}\right) - E \left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z - z}{h_n}\right) \right] \right) \\
&- \frac{1}{n \cdot h_n^{dz}} \sum_{i=1}^n \left(\frac{\partial^2 \mu_S^{NC}(X_i, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right) - E \left[\frac{\partial^2 \mu_S^{NC}(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X, \theta) \cdot K\left(\frac{Z - z}{h_n}\right) \right] \right) \\
&- \frac{1}{n \cdot h_n^{dz}} \sum_{i=1}^n \left(\frac{\partial \mu_S^{NC}(X_i, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z_i - z}{h_n}\right) - E \left[\frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z - z}{h_n}\right) \right] \right) \\
&+ \left(E \left[\frac{1}{h_n^{dz}} \cdot \frac{\partial^2 \Xi_S(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot (S - \mu_S^{NC}(X, \theta)) \cdot K\left(\frac{Z - z}{h_n}\right) \right] - \delta_{I, \theta_\ell, \theta_j}(z, \theta) \cdot f_Z(z) \right) \\
&- \left(E \left[\frac{1}{h_n^{dz}} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z - z}{h_n}\right) \right] - \delta_{II, \theta_\ell, \theta_j}(z, \theta) \cdot f_Z(z) \right) \\
&- \left(E \left[\frac{1}{h_n^{dz}} \cdot \frac{\partial^2 \mu_S^{NC}(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X, \theta) \cdot K\left(\frac{Z - z}{h_n}\right) \right] - \delta_{III, \theta_\ell, \theta_j}(z, \theta) \cdot f_Z(z) \right) \\
&- \left(E \left[\frac{1}{h_n^{dz}} \cdot \frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z - z}{h_n}\right) \right] - \delta_{IV, \theta_j, \theta_\ell}(z, \theta) \cdot f_Z(z) \right)
\end{aligned} \tag{S1.29}$$

and,

$$\begin{aligned}
\frac{\partial^2 \widehat{R}_b^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} &= \frac{\partial^2 R_b^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} \\
&+ \frac{2}{n \cdot h_n^{dz}} \sum_{i=1}^n \left(\frac{\partial^2 \Xi_S(X_i, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right) - E \left[\frac{\partial^2 \Xi_S(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X, \theta) \cdot K\left(\frac{Z - z}{h_n}\right) \right] \right) \\
&+ \frac{2}{n \cdot h_n^{dz}} \sum_{i=1}^n \left(\frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z_i - z}{h_n}\right) - E \left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z - z}{h_n}\right) \right] \right) \\
&+ 2 \cdot \left(E \left[\frac{1}{h_n^{dz}} \cdot \frac{\partial^2 \Xi_S(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X, \theta) \cdot K\left(\frac{Z - z}{h_n}\right) \right] - \delta_{IV, \theta_\ell, \theta_j}(z, \theta) \cdot f_Z(z) \right) \\
&+ 2 \cdot \left(E \left[\frac{1}{h_n^{dz}} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z - z}{h_n}\right) \right] - \delta_{V, \theta_\ell, \theta_j}(z, \theta) \cdot f_Z(z) \right)
\end{aligned} \tag{S1.30}$$

By the bounded-variation properties of our kernel function described in Assumption E1 and the Lipschitz-restrictions in Assumption E3, Nolan and Pollard (1987, Lemma 22), Pakes and Pollard (1989, Example 2.10, Lemma 2.13, Lemma 2.14) imply that the classes of functions that produce the empirical processes described next are Euclidean. The existence of $2 + \delta$ -moments in Assumptions I1 and E3, and the boundedness restrictions in part (iii) of Assumption G1 imply that the corresponding envelopes for these classes also

have finite $2 + \delta$ -moments. From here, Sherman (1994, Corollary 4) can be used to show that,

$$\begin{aligned}
& \sup_{\substack{\theta \in \Theta \\ z \in Z}} \left| \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \left(\frac{\partial^2 \Xi_S(X_i, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot (S_i - \mu_S^{NC}(X_i, \theta)) \cdot K\left(\frac{Z_i - z}{h_n}\right) - E \left[\frac{\partial^2 \Xi_S(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot (S - \mu_S^{NC}(X, \theta)) \cdot K\left(\frac{Z - z}{h_n}\right) \right] \right) \right| = o_p(1), \\
& \sup_{\substack{\theta \in \Theta \\ z \in Z}} \left| \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \left(\frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \mu_S^{NC}(X_i, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z_i - z}{h_n}\right) - E \left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z - z}{h_n}\right) \right] \right) \right| = o_p(1), \\
& \sup_{\substack{\theta \in \Theta \\ z \in Z}} \left| \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \left(\frac{\partial^2 \mu_S^{NC}(X_i, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right) - E \left[\frac{\partial^2 \mu_S^{NC}(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X, \theta) \cdot K\left(\frac{Z - z}{h_n}\right) \right] \right) \right| = o_p(1), \\
& \sup_{\substack{\theta \in \Theta \\ z \in Z}} \left| \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \left(\frac{\partial \mu_S^{NC}(X_i, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z_i - z}{h_n}\right) - E \left[\frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z - z}{h_n}\right) \right] \right) \right| = o_p(1), \\
& \sup_{\substack{\theta \in \Theta \\ z \in Z}} \left| \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \left(\frac{\partial^2 \Xi_S(X_i, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right) - E \left[\frac{\partial^2 \Xi_S(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X, \theta) \cdot K\left(\frac{Z - z}{h_n}\right) \right] \right) \right| = o_p(1), \\
& \sup_{\substack{\theta \in \Theta \\ z \in Z}} \left| \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \left(\frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z_i - z}{h_n}\right) - E \left[\frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z - z}{h_n}\right) \right] \right) \right| = o_p(1)
\end{aligned}$$

And for the bias terms, the smoothness restrictions in Assumption E2 yield,

$$\begin{aligned}
& \sup_{\substack{\theta \in \Theta \\ z \in Z}} \left| E \left[\frac{1}{h_n^{d_Z}} \cdot \frac{\partial^2 \Xi_S(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot (S - \mu_S^{NC}(X, \theta)) \cdot K\left(\frac{Z - z}{h_n}\right) \right] - \delta_{I, \theta_\ell, \theta_j}(z, \theta) \cdot f_Z(z) \right| = o(1), \\
& \sup_{\substack{\theta \in \Theta \\ z \in Z}} \left| E \left[\frac{1}{h_n^{d_Z}} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z - z}{h_n}\right) \right] - \delta_{II, \theta_\ell, \theta_j}(z, \theta) \cdot f_Z(z) \right| = o(1), \\
& \sup_{\substack{\theta \in \Theta \\ z \in Z}} \left| E \left[\frac{1}{h_n^{d_Z}} \cdot \frac{\partial^2 \mu_S^{NC}(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X, \theta) \cdot K\left(\frac{Z - z}{h_n}\right) \right] - \delta_{III, \theta_\ell, \theta_j}(z, \theta) \cdot f_Z(z) \right| = o(1), \\
& \sup_{\substack{\theta \in \Theta \\ z \in Z}} \left| E \left[\frac{1}{h_n^{d_Z}} \cdot \frac{\partial \mu_S^{NC}(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z - z}{h_n}\right) \right] - \delta_{IV, \theta_j, \theta_\ell}(z, \theta) \cdot f_Z(z) \right| = o(1), \\
& \sup_{\substack{\theta \in \Theta \\ z \in Z}} \left| E \left[\frac{1}{h_n^{d_Z}} \cdot \frac{\partial^2 \Xi_S(X, \theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_S(X, \theta) \cdot K\left(\frac{Z - z}{h_n}\right) \right] - \delta_{IV, \theta_\ell, \theta_j}(z, \theta) \cdot f_Z(z) \right| = o(1), \\
& \sup_{\substack{\theta \in \Theta \\ z \in Z}} \left| E \left[\frac{1}{h_n^{d_Z}} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_S(X, \theta)}{\partial \theta_j} \cdot K\left(\frac{Z - z}{h_n}\right) \right] - \delta_{V, \theta_\ell, \theta_j}(z, \theta) \cdot f_Z(z) \right| = o(1).
\end{aligned}$$

Combined with equations (S1.29)-(S1.30), these results yield,

$$\begin{aligned} \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial^2 \widehat{R}_a^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} - \frac{\partial^2 R_a^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} \right| &= o_p(1), \quad \ell, j = 1, \dots, d_\theta, \\ \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial^2 \widehat{R}_b^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} - \frac{\partial^2 R_b^\pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} \right| &= o_p(1), \quad \ell, j = 1, \dots, d_\theta. \end{aligned} \quad (\text{S1.31})$$

From the expressions in (S1.25) and (S1.26), the results summarized in (S1.27), combined with (S1.31) yield,

$$\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial^2 \widehat{\pi}(z, \theta)}{\partial \theta_j \partial \theta_\ell} - \frac{\partial^2 \pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} \right| = o_p(1) \text{ for } j, \ell = 1, \dots, d_\theta.$$

To summarize, our restrictions yield the linear representation results,

$$\begin{aligned} \widehat{\pi}(z, \theta) &= \pi(z, \theta) + \frac{1}{n \cdot h_n^{d_z}} \sum_{i=1}^n \psi_n^\pi(V_i; z, \theta) + \mathfrak{V}_n^\pi(z, \theta), \\ \text{where } \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\mathfrak{V}_n^\pi(z, \theta)| &= o_p(n^{-1/2}), \\ \frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_\ell} &= \frac{\partial \pi(z, \theta)}{\partial \theta_\ell} + \frac{1}{n \cdot h_n^{d_z}} \sum_{i=1}^n \psi_n^{\nabla \theta_\ell \pi}(V_i; z, \theta) + \mathfrak{V}_n^{\nabla \theta_\ell \pi}(z, \theta), \\ \text{where } \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\mathfrak{V}_n^{\nabla \theta_\ell \pi}(z, \theta)| &= o_p(n^{-1/2}), \quad \ell = 1, \dots, d_\theta, \end{aligned}$$

where the influence function $\psi_n^\pi(V_i; z, \theta)$ is as described in equation (S1.10), and the influence function $\psi_n^{\nabla \theta_\ell \pi}(V_i; z, \theta)$ is as described in equation (S1.22). Finally,

$$\begin{aligned} \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\widehat{\pi}(z, \theta) - \pi(z, \theta)| &= o_p(n^{-1/4}), \\ \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial \widehat{\pi}(z, \theta)}{\partial \theta_\ell} - \frac{\partial \pi(z, \theta)}{\partial \theta_\ell} \right| &= o_p(n^{-1/4}), \quad \ell = 1, \dots, d_\theta, \\ \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial^2 \widehat{\pi}(z, \theta)}{\partial \theta_j \partial \theta_\ell} - \frac{\partial^2 \pi(z, \theta)}{\partial \theta_j \partial \theta_\ell} \right| &= o_p(1), \quad j, \ell = 1, \dots, d_\theta. \end{aligned} \quad (\text{S1.32})$$

S1.2 Asymptotic properties of $|\widehat{Q}_{S,\mathcal{Z}}(\theta) - Q_{S,\mathcal{Z}}(\theta)|$

Recall that the population objective function of our conditional GMM estimator is

$$Q_{S,\mathcal{Z}}(\theta) \equiv \frac{1}{2} \cdot E[\tau_{\mathcal{Z}}(U, \theta)^2], \quad \text{where} \quad \tau_{\mathcal{Z}}(u, \theta) \equiv E[\varphi_S(V, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U \leq u\}],$$

and $\varphi_S(V_i, \theta) \equiv S_i - \mu_S^{NC}(X_i, \theta) - \pi(Z_i, \theta) \cdot \Xi_S(X_i, \theta)$. Our sample objective function is,

$$\widehat{Q}_{S,\mathcal{Z}}(\theta) \equiv \frac{1}{2} \cdot \frac{1}{n} \sum_{j=1}^n \widehat{\tau}_{\mathcal{Z}}(U_j, \theta)^2 \quad (\text{S1.33})$$

where,

$$\begin{aligned} \widehat{\tau}_{\mathcal{Z}}(u, \theta) &= \frac{1}{n-1} \sum_{i \neq j} \widehat{\varphi}_S(V_i, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq u\}, \\ \widehat{\varphi}_S(V_i, \theta) &\equiv S_i - \mu_S^{NC}(X_i, \theta) - \widehat{\pi}(Z_i, \theta) \cdot \Xi_S(X_i, \theta) \end{aligned}$$

We have,

$$\begin{aligned} \widehat{\tau}_{\mathcal{Z}}(u, \theta) - \frac{1}{n} \sum_{i=1}^n \varphi_S(V_i, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq u\} &= \frac{1}{n} \sum_{i=1}^n (\widehat{\varphi}_S(V_i, \theta) - \varphi_S(V_i, \theta)) \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq u\} \\ &= -\frac{1}{n} \sum_{i=1}^n [\widehat{\pi}(Z_i, \theta) - \pi(Z_i, \theta)] \cdot \Xi_S(X_i, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq u\} \end{aligned}$$

Note that $|\Xi_S(x, \theta)| \leq 1 \ \forall (x, \theta)$, and $\mathbb{1}_{\mathcal{Z}}\{U_i \leq u\} \leq \mathbb{1}\{Z_i \in \mathcal{Z}\} \ \forall u$. Thus,

$$\sup_{\substack{\theta \in \Theta \\ u \in \mathbb{R}^{d_U}}} \left| \widehat{\tau}_{\mathcal{Z}}(u, \theta) - \frac{1}{n} \sum_{i=1}^n \varphi_S(V_i, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq u\} \right| \leq \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\widehat{\pi}(z, \theta) - \pi(z, \theta)| = o_p(1) \quad (\text{S1.34})$$

Given the Lipschitz restrictions in Assumption E3, Pakes and Pollard (1989, Lemmas 2.12, 2.13 and 2.14)) imply that the class of functions $\{\varphi_S(V, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{u \leq c\} \text{ for some } \theta \in \Theta, c \in \mathbb{R}^{d_U}\}$ is a *Euclidean* class (see Sherman (1994, Definition 3)), and given the existence of $2 + \delta$ -moments in Assumptions I1 and E3, and the regularity properties of the parametric distribution described in Assumption G1, the corresponding envelope for the class also has finite $2 + \delta$ -moments. From here, Sherman (1994, Corollary 4) yields,

$$\sup_{\substack{\theta \in \Theta \\ u \in \mathbb{R}^{d_U}}} \left| \frac{1}{n} \sum_{i=1}^n (\varphi_S(V_i, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq u\} - \tau_{\mathcal{Z}}(u, \theta)) \right| = o_p(1).$$

Combined with the result in (S1.34), this yields,

$$\sup_{\substack{\theta \in \Theta \\ u \in \mathbb{R}^d U}} |\widehat{\tau}_{\mathcal{Z}}(u, \theta) - \tau_{\mathcal{Z}}(u, \theta)| = o_p(1). \quad (\text{S1.35})$$

Plugging (S1.35) into (S1.33),

$$\widehat{Q}_{S, \mathcal{Z}}(\theta) = \frac{1}{2} \cdot \frac{1}{n} \sum_{j=1}^n \tau_{\mathcal{Z}}(U_j, \theta)^2 + \xi_{I, n}^{\widehat{Q}}(\theta), \quad \text{Where} \quad \sup_{\theta \in \Theta} |\xi_{I, n}^{\widehat{Q}}(\theta)| = o_p(1). \quad (\text{S1.36})$$

Recall that $Q_{S, \mathcal{Z}}(\theta) \equiv \frac{1}{2} \cdot E[\tau_{\mathcal{Z}}(U, \theta)^2]$. Given the Lipschitz restrictions in Assumption E3, Pakes and Pollard (1989, Lemmas 2.13 and 2.14)) imply that the class of functions $\{\tau(u, \theta) \text{ for some } \theta \in \Theta\}$ is Euclidean, and the regularity properties of the parametric distribution described in Assumption G1, the corresponding envelope for the class also has finite $2 + \delta$ -moments. From here, Sherman (1994, Corollary 4) yields,

$$\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{j=1}^n \tau_{\mathcal{Z}}(U_j, \theta)^2 - E[\tau_{\mathcal{Z}}(U, \theta)^2] \right| = o_p(1). \quad (\text{S1.37})$$

Plugging (S1.37) into (S1.36), we obtain

$$\sup_{\theta \in \Theta} |\widehat{Q}_{S, \mathcal{Z}}(\theta) - Q_{S, \mathcal{Z}}(\theta)| = o_p(1). \quad (\text{S1.38})$$

S1.3 Asymptotic properties of $\left| \frac{\partial^2 \widehat{Q}_{S, \mathcal{Z}}(\theta)}{\partial \theta \partial \theta'} - \frac{\partial^2 Q_{S, \mathcal{Z}}(\theta)}{\partial \theta \partial \theta'} \right|$

We have

$$\begin{aligned} \frac{\partial^2 Q_{S, \mathcal{Z}}(\theta)}{\partial \theta \partial \theta'} &= E \left[\frac{\partial^2 \tau_{\mathcal{Z}}(U, \theta)}{\partial \theta \partial \theta'} \cdot \tau_{\mathcal{Z}}(U, \theta) \right] + E \left[\frac{\partial \tau_{\mathcal{Z}}(U, \theta)}{\partial \theta} \cdot \frac{\partial \tau_{\mathcal{Z}}(U, \theta)}{\partial \theta} \right], \\ \frac{\partial^2 \widehat{Q}_{S, \mathcal{Z}}(\theta)}{\partial \theta \partial \theta'} &= \frac{1}{n} \sum_{j=1}^n \left(\frac{\partial^2 \widehat{\tau}_{\mathcal{Z}}(U_j, \theta)}{\partial \theta \partial \theta'} \cdot \widehat{\tau}_{\mathcal{Z}}(U_j, \theta) + \frac{\partial \widehat{\tau}_{\mathcal{Z}}(U_j, \theta)}{\partial \theta} \cdot \frac{\partial \widehat{\tau}_{\mathcal{Z}}(U_j, \theta)}{\partial \theta} \right). \end{aligned} \quad (\text{S1.39})$$

Where,

$$\begin{aligned}\frac{\partial \tau_Z(u, \theta)}{\partial \theta} &= E \left[\frac{\partial \varphi_S(V, \theta)}{\partial \theta} \cdot \mathbb{1}_Z\{U \leq u\} \right], \\ \frac{\partial^2 \tau_Z(u, \theta)}{\partial \theta \partial \theta'} &= E \left[\frac{\partial^2 \varphi_S(V, \theta)}{\partial \theta \partial \theta'} \cdot \mathbb{1}_Z\{U \leq u\} \right], \\ \frac{\partial \widehat{\tau}_Z(u, \theta)}{\partial \theta} &= \frac{1}{n} \sum_{i=1}^n \frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \theta} \cdot \mathbb{1}_Z\{U_i \leq u\}, \\ \frac{\partial^2 \widehat{\tau}_Z(u, \theta)}{\partial \theta \partial \theta'} &= \frac{1}{n} \sum_{i=1}^n \frac{\partial^2 \widehat{\varphi}_S(V_i, \theta)}{\partial \theta \partial \theta'} \cdot \mathbb{1}_Z\{U_i \leq u\}.\end{aligned}$$

Recall that,

$$\begin{aligned}\widehat{\varphi}_S(V_i, \theta) &\equiv S_i - \mu_S^{NC}(X_i, \theta) - \widehat{\pi}(Z_i, \theta) \cdot \Xi_S(X_i, \theta), \\ \varphi_S(V_i, \theta) &\equiv S_i - \mu_S^{NC}(X_i, \theta) - \pi(Z_i, \theta) \cdot \Xi_S(X_i, \theta).\end{aligned}$$

Thus,

$$\begin{aligned}\frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \theta} - \frac{\partial \varphi_S(V_i, \theta)}{\partial \theta} &= \left[\frac{\partial \pi(Z_i, \theta)}{\partial \theta} - \frac{\partial \widehat{\pi}(Z_i, \theta)}{\partial \theta} \right] \cdot \Xi_S(X_i, \theta) \\ &\quad + \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta} \cdot [\pi(Z_i, \theta) - \widehat{\pi}(Z_i, \theta)],\end{aligned}\tag{S1.40}$$

and,

$$\begin{aligned}\frac{\partial^2 \widehat{\varphi}_S(V_i, \theta)}{\partial \theta \partial \theta'} - \frac{\partial^2 \varphi_S(V_i, \theta)}{\partial \theta \partial \theta'} &= \left[\frac{\partial \pi(Z_i, \theta)}{\partial \theta} - \frac{\partial \widehat{\pi}(Z_i, \theta)}{\partial \theta} \right] \cdot \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta}' \\ &\quad + \left[\frac{\partial^2 \pi(Z_i, \theta)}{\partial \theta \partial \theta'} - \frac{\partial^2 \widehat{\pi}(Z_i, \theta)}{\partial \theta \partial \theta'} \right] \cdot \Xi_S(X_i, \theta) + \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta} \cdot \left[\frac{\partial \pi(Z_i, \theta)}{\partial \theta} - \frac{\partial \widehat{\pi}(Z_i, \theta)}{\partial \theta} \right]' \\ &\quad + \frac{\partial^2 \Xi_S(X_i, \theta)}{\partial \theta \partial \theta'} \cdot [\pi(Z_i, \theta) - \widehat{\pi}(Z_i, \theta)].\end{aligned}\tag{S1.41}$$

Under the existence-of-moments restriction in Assumption I1, and the regularity conditions in Assumption E3, we have

$$\sup_{\theta \in \Theta} \left(\frac{1}{n} \sum_{i=1}^n \left\| \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta} \right\| \right) = O_p(1), \quad \text{and} \quad \sup_{\theta \in \Theta} \left(\frac{1}{n} \sum_{i=1}^n \left\| \frac{\partial^2 \Xi_S(X_i, \theta)}{\partial \theta \partial \theta'} \right\| \right) = O_p(1).$$

Furthermore, $|\Xi_S(x, \theta)| \leq 1$ for all (x, θ) . Thus, from the expressions in (S1.40)-(S1.41), the uniform convergence results in (S1.32)-(S1.32) yield

$$\begin{aligned} \sup_{\substack{\theta \in \Theta \\ u \in \mathbb{R}^{d_U}}} \left| \frac{\partial \widehat{\tau}_Z(u, \theta)}{\partial \theta} - \frac{1}{n-1} \sum_{i \neq j} \frac{\partial \varphi_S(V_i, \theta)}{\partial \theta} \cdot \mathbb{1}_Z\{U_i \leq u\} \right| &= o_p(1), \\ \sup_{\substack{\theta \in \Theta \\ u \in \mathbb{R}^{d_U}}} \left| \frac{\partial^2 \widehat{\tau}_Z(u, \theta)}{\partial \theta \partial \theta'} - \frac{1}{n-1} \sum_{i \neq j} \frac{\partial^2 \varphi_S(V_i, \theta)}{\partial \theta \partial \theta'} \cdot \mathbb{1}_Z\{U_i \leq u\} \right| &= o_p(1), \end{aligned} \quad (\text{S1.42})$$

Next, given the Lipschitz restrictions in Assumption E3, Pakes and Pollard (1989, Lemmas 2.12, 2.13 and 2.14)) imply that the class of functions

$$\begin{aligned} &\left\{ \frac{\partial \varphi_S(V, \theta)}{\partial \theta} \cdot \mathbb{1}_Z\{u \leq c\} \text{ for some } \theta \in \Theta, c \in \mathbb{R}^{d_U} \right\}, \\ &\left\{ \frac{\partial^2 \varphi_S(V, \theta)}{\partial \theta \partial \theta'} \cdot \mathbb{1}_Z\{u \leq c\} \text{ for some } \theta \in \Theta, c \in \mathbb{R}^{d_U} \right\} \end{aligned}$$

are Euclidean. The regularity properties of the parametric distribution described in Assumption G1, and the existence of $2 + \delta$ -moment restrictions described in Assumption I1 imply that the corresponding envelope for the class also has finite $2 + \delta$ -moments. From here, Sherman (1994, Corollary 4) yields,

$$\begin{aligned} \sup_{\substack{\theta \in \Theta \\ u \in \mathbb{R}^{d_U}}} \left| \frac{1}{n} \sum_{i=1}^n \left(\frac{\partial \varphi_S(V_i, \theta)}{\partial \theta} \cdot \mathbb{1}_Z\{U_i \leq u\} - \frac{\partial \tau_Z(u, \theta)}{\partial \theta} \right) \right| &= o_p(1), \\ \sup_{\substack{\theta \in \Theta \\ u \in \mathbb{R}^{d_U}}} \left| \frac{1}{n} \sum_{i=1}^n \left(\frac{\partial^2 \varphi_S(V_i, \theta)}{\partial \theta \partial \theta'} \cdot \mathbb{1}_Z\{U_i \leq u\} - \frac{\partial^2 \tau_Z(u, \theta)}{\partial \theta \partial \theta'} \right) \right| &= o_p(1). \end{aligned}$$

Combined with the results in (S1.42), this yields,

$$\begin{aligned} \sup_{\substack{\theta \in \Theta \\ u \in \mathbb{R}^{d_U}}} \left| \frac{\partial \widehat{\tau}_Z(u, \theta)}{\partial \theta} - \frac{\partial \tau_Z(u, \theta)}{\partial \theta} \right| &= o_p(1), \\ \sup_{\substack{\theta \in \Theta \\ u \in \mathbb{R}^{d_U}}} \left| \frac{\partial^2 \widehat{\tau}_Z(u, \theta)}{\partial \theta \partial \theta'} - \frac{\partial^2 \tau_Z(u, \theta)}{\partial \theta \partial \theta'} \right| &= o_p(1) \end{aligned}$$

Combining this result with (S1.35) and plugging back into (S1.39),

$$\frac{\partial^2 \widehat{Q}_{S,Z}(\theta)}{\partial \theta \partial \theta'} = \frac{1}{n} \sum_{j=1}^n \left(\frac{\partial^2 \tau_Z(U_j, \theta)}{\partial \theta \partial \theta'} \cdot \tau_Z(U_j, \theta) + \frac{\partial \tau_Z(U_j, \theta)}{\partial \theta} \cdot \frac{\partial \tau_Z(U_j, \theta)'}{\partial \theta} \right) + \xi_{I,n}^{\nabla_{\theta\theta'}} \widehat{Q}(\theta), \quad (\text{S1.43})$$

where $\sup_{\theta \in \Theta} \left| \xi_{I,n}^{\nabla_{\theta\theta'}} \widehat{Q}(\theta) \right| = o_p(1).$

Finally, the Lipschitz restrictions in Assumption E3, Pakes and Pollard (1989, Lemmas 2.12, 2.13 and 2.14)) imply that the following class of functions is Euclidean (element-wise),

$$\left\{ \frac{\partial^2 \tau(u, \theta)}{\partial \theta \partial \theta'} \cdot \tau_Z(u, \theta) + \frac{\partial \tau_Z(u, \theta)}{\partial \theta} \cdot \frac{\partial \tau_Z(u, \theta)'}{\partial \theta} \text{ for some } \theta \in \Theta \right\}$$

The regularity properties of the parametric distribution described in Assumption G1, and the existence of $2+\delta$ -moment restrictions described in Assumption I1 imply that the corresponding envelope for the class also has finite $2+\delta$ -moments. From here, Sherman (1994, Corollary 4) and the definition of $\frac{\partial^2 Q_{S,Z}(\theta)}{\partial \theta \partial \theta'}$ in (S1.39) yield,

$$\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{j=1}^n \left[\left(\frac{\partial^2 \tau_Z(U_j, \theta)}{\partial \theta \partial \theta'} \cdot \tau_Z(U_j, \theta) + \frac{\partial \tau_Z(U_j, \theta)}{\partial \theta} \cdot \frac{\partial \tau_Z(U_j, \theta)'}{\partial \theta} \right) - \frac{\partial^2 Q_{S,Z}(\theta)}{\partial \theta \partial \theta'} \right] \right| = o_p(1)$$

Plugging this back into (S1.43), we finally obtain,

$$\sup_{\theta \in \Theta} \left| \frac{\partial^2 \widehat{Q}_{S,Z}(\theta)}{\partial \theta \partial \theta'} - \frac{\partial^2 Q_{S,Z}(\theta)}{\partial \theta \partial \theta'} \right| = o_p(1). \quad (\text{S1.44})$$

S1.4 Consistency

Under the restrictions of Assumptions G1, G2 and I1, our estimator first-step MLE estimator $\widehat{\gamma}$ satisfies $\widehat{\gamma} \xrightarrow{p} \gamma_0$. Next, given the restrictions in Assumption I4, θ_0 is the unique minimizer of $Q_{S,Z}(\theta)$ over Θ . Furthermore, $Q_{S,Z}(\theta)$ is continuous by the restrictions in Assumption I3. Also, as we showed in (S1.38), under our restrictions we have $\sup_{\theta \in \Theta} |\widehat{Q}_{S,Z}(\theta) - Q_{S,Z}(\theta)| = o_p(1)$. Thus, since $\widehat{\gamma} \xrightarrow{p} \gamma_0$, we have, $\sup_{\Delta \in \Theta} |\widehat{Q}_{S,Z}(\widehat{\gamma}, \Delta) - Q_{S,Z}(\gamma_0, \Delta)| = o_p(1)$. From here, Newey and McFadden (1994, Theorem 2.1) yields $\widehat{\Delta} \xrightarrow{p} \Delta_0$. Thus, grouping $\widehat{\theta} \equiv (\widehat{\gamma}', \widehat{\Delta}')'$, we have $\widehat{\theta} \xrightarrow{p} \theta_0$.

S1.5 Asymptotic distribution

Since $\theta_0 \in \text{int}(\Theta)$ by assumption, smoothness of our sample objective function $\widehat{Q}_{S,Z}(\theta)$ implies that, with probability approaching one (w.p.a.1), $\widehat{\theta}$ satisfies the first-order conditions

$\frac{\partial \widehat{Q}_{S,Z}(\widehat{\theta})}{\partial \Delta} = 0$. From here, a linear approximation yields,

$$0 = \frac{\partial \widehat{Q}_{S,Z}(\widehat{\theta})}{\partial \Delta} = \frac{\partial \widehat{Q}_{S,Z}(\theta_0)}{\partial \Delta} + \frac{\partial^2 \widehat{Q}_{S,Z}(\widetilde{\theta})}{\partial \Delta \partial \gamma'} \cdot (\widehat{\gamma} - \gamma_0) + \frac{\partial^2 \widehat{Q}_{S,Z}(\widetilde{\theta})}{\partial \Delta \partial \Delta'} \cdot (\widehat{\Delta} - \Delta_0)$$

As usual, $\widetilde{\theta}$ belongs in the line segment connecting $\widehat{\theta}$ and θ_0 and, since $\widehat{\theta} \xrightarrow{p} \theta_0$, it satisfies $\widetilde{\theta} \xrightarrow{p} \theta_0$. Given the uniform convergence result $\sup_{\theta \in \Theta} \left| \frac{\partial^2 \widehat{Q}_{S,Z}(\theta)}{\partial \theta \partial \theta'} - \frac{\partial^2 Q_{S,Z}(\theta)}{\partial \theta \partial \theta'} \right| = o_p(1)$ in equation (S1.44) and the assumption that $\frac{\partial^2 Q_{S,Z}(\theta)}{\partial \Delta \partial \Delta'}$ exists and is a continuous function of θ in an open neighborhood $\mathcal{N}$ of θ_0 (see Assumption I3), it follows that with probability approaching one, we can express,

$$\widehat{\Delta} - \Delta_0 = - \left(\frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \Delta \partial \Delta'}^{-1} + o_p(1) \right) \times \left[\frac{\partial \widehat{Q}_{S,Z}(\theta_0)}{\partial \Delta} + \left(\frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \Delta \partial \gamma'} + o_p(1) \right) \cdot (\widehat{\gamma} - \gamma_0) \right] \quad (\text{S1.45})$$

Plugging in the linear representation result of our first-stage estimator $\widehat{\gamma}$, we obtain,

$$\widehat{\Delta} - \Delta_0 = - \left(\frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \Delta \partial \Delta'}^{-1} + o_p(1) \right) \times \left[\frac{\partial \widehat{Q}_{S,Z}(\theta_0)}{\partial \Delta} + \left(\frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \Delta \partial \gamma'} + o_p(1) \right) \cdot \left(\frac{1}{n} \sum_{i=1}^n \psi_{\gamma}(V_i; \gamma_0) + o_p(n^{-1/2}) \right) \right] \quad (\text{S1.46})$$

Our next step is to show that we can obtain a linear representation result for $\frac{\partial \widehat{Q}_{S,Z}(\theta_0)}{\partial \Delta}$.

S1.5.1 A linear representation for $\frac{\partial \widehat{Q}_{S,Z}(\theta_0)}{\partial \Delta}$

We have,

$$\begin{aligned} \frac{\partial \widehat{Q}_{S,Z}(\theta)}{\partial \Delta} &= \frac{1}{n} \sum_{j=1}^n \left[\left(\frac{1}{n} \sum_{i=1}^n \widehat{\varphi}_S(V_i, \theta) \cdot \mathbb{1}_{\mathcal{Z}\{U_i \leq U_j\}} \right) \times \left(\frac{1}{n} \sum_{\ell=1}^n \frac{\partial \widehat{\varphi}_S(V_\ell, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}\{U_\ell \leq U_j\}} \right) \right] \\ &= \frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i,j} \widehat{\varphi}_S(V_i, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_\ell, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}\{U_i \leq U_j\}} \cdot \mathbb{1}_{\mathcal{Z}\{U_\ell \leq U_j\}} \\ &\quad + \frac{1}{n} \times \left[\frac{1}{n^2} \sum_{j=1}^n \sum_{i \neq j} \left(\widehat{\varphi}_S(V_i, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}\{U_i \leq U_j\}} + \widehat{\varphi}_S(V_j, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}\{U_i \leq U_j\}} \right. \right. \\ &\quad \left. \left. + \widehat{\varphi}_S(V_i, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_j, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}\{U_i \leq U_j\}} \right) \right] \\ &\quad + \frac{1}{n^2} \times \left[\frac{1}{n} \sum_{j=1}^n \widehat{\varphi}_S(V_j, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_j, \theta)}{\partial \Delta} \right] \end{aligned}$$

Writing $\widehat{\varphi}_S(V_i, \theta) = \varphi(V_i, \theta) + [\widehat{\varphi}_S(V_i, \theta) - \varphi(V_i, \theta)]$ and $\frac{\partial \widehat{\varphi}_S(V_\ell, \theta)}{\partial \Delta} = \frac{\partial \varphi(V_\ell, \theta)}{\partial \Delta} + \left[\frac{\partial \widehat{\varphi}_S(V_\ell, \theta)}{\partial \Delta} - \frac{\partial \varphi(V_\ell, \theta)}{\partial \Delta} \right]$, we obtain,

$$\begin{aligned}
\frac{\partial \widehat{Q}_{S,Z}(\theta)}{\partial \Delta} = & \frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} \varphi_S(V_i, \theta) \cdot \frac{\partial \varphi_S(V_\ell, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}\{U_i \leq U_j\}} \cdot \mathbb{1}_{\mathcal{Z}\{U_\ell \leq U_j\}} \\
& + \frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} \varphi_S(V_i, \theta) \cdot \left[\frac{\partial \widehat{\varphi}_S(V_\ell, \theta)}{\partial \Delta} - \frac{\partial \varphi_S(V_\ell, \theta)}{\partial \Delta} \right] \cdot \mathbb{1}_{\mathcal{Z}\{U_i \leq U_j\}} \cdot \mathbb{1}_{\mathcal{Z}\{U_\ell \leq U_j\}} \\
& + \frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} \frac{\partial \varphi_S(V_\ell, \theta)}{\partial \Delta} \cdot [\widehat{\varphi}_S(V_i, \theta) - \varphi_S(V_i, \theta)] \cdot \mathbb{1}_{\mathcal{Z}\{U_i \leq U_j\}} \cdot \mathbb{1}_{\mathcal{Z}\{U_\ell \leq U_j\}} \\
& + \frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} [\widehat{\varphi}_S(V_i, \theta) - \varphi_S(V_i, \theta)] \cdot \left[\frac{\partial \widehat{\varphi}_S(V_\ell, \theta)}{\partial \Delta} - \frac{\partial \varphi_S(V_\ell, \theta)}{\partial \Delta} \right] \cdot \mathbb{1}_{\mathcal{Z}\{U_i \leq U_j\}} \cdot \mathbb{1}_{\mathcal{Z}\{U_\ell \leq U_j\}} \\
& + \frac{1}{n} \times \left[\frac{1}{n^2} \sum_{j=1}^n \sum_{i \neq j} \left(\widehat{\varphi}_S(V_i, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}\{U_i \leq U_j\}} + \widehat{\varphi}_S(V_j, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}\{U_i \leq U_j\}} \right. \right. \\
& \quad \left. \left. + \widehat{\varphi}_S(V_i, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_j, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}\{U_i \leq U_j\}} \right) \right] \\
& + \frac{1}{n^2} \times \left[\frac{1}{n} \sum_{j=1}^n \widehat{\varphi}_S(V_j, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_j, \theta)}{\partial \Delta} \right]
\end{aligned} \tag{S1.47}$$

Recall that, $\widehat{\varphi}_S(V_i, \theta) \equiv S_i - \mu_S^{NC}(X_i, \theta) - \widehat{\pi}(Z_i, \theta) \cdot \Xi_S(X_i, \theta)$, and $\varphi_S(V_i, \theta) \equiv S_i - \mu_S^{NC}(X_i, \theta) - \pi(Z_i, \theta) \cdot \Xi_S(X_i, \theta)$. Thus,

$$\begin{aligned}
\widehat{\varphi}_S(V_i, \theta) - \varphi_S(V_i, \theta) &= -[\widehat{\pi}(Z_i, \theta) - \pi(Z_i, \theta)] \cdot \Xi_S(X_i, \theta), \\
\frac{\partial \widehat{\varphi}_S(V_\ell, \theta)}{\partial \Delta} - \frac{\partial \varphi_S(V_\ell, \theta)}{\partial \Delta} &= -\left[\frac{\partial \widehat{\pi}(Z_\ell, \theta)}{\partial \Delta} - \frac{\partial \pi(Z_\ell, \theta)}{\partial \Delta} \right] \cdot \Xi_S(X_\ell, \theta) - [\widehat{\pi}(Z_\ell, \theta) - \pi(Z_\ell, \theta)] \cdot \frac{\partial \Xi_S(X_\ell, \theta)}{\partial \Delta}
\end{aligned} \tag{S1.48}$$

Let the influence functions $\psi_n^\pi(v; z, \theta)$ and $\psi_n^{\nabla \Delta \pi}(v; z, \theta)$ be as described in equations (S1.10) and (S1.22), and let $\vartheta_n^\pi(z, \theta)$ and $\vartheta_n^{\nabla \Delta \pi}(z, \theta)$ be the remainders of the linear representations described in (S1.10) and (S1.23). Plugging in these linear representations into (S1.48),

$$\begin{aligned}
\widehat{\varphi}_S(V_i, \theta) - \varphi_S(V_i, \theta) &= -\left(\frac{1}{n \cdot h_n^{d_Z}} \sum_{k=1}^n \psi_n^\pi(V_k; Z_i, \theta) + \vartheta_n^\pi(Z_i, \theta) \right) \cdot \Xi_S(X_i, \theta), \\
\frac{\partial \widehat{\varphi}_S(V_\ell, \theta)}{\partial \Delta} - \frac{\partial \varphi_S(V_\ell, \theta)}{\partial \Delta} &= -\left(\frac{1}{n \cdot h_n^{d_Z}} \sum_{k=1}^n \psi_n^{\nabla \Delta \pi}(V_k; Z_\ell, \theta) + \vartheta_n^{\nabla \Delta \pi}(Z_\ell, \theta) \right) \cdot \Xi_S(X_\ell, \theta) \\
&\quad - \left(\frac{1}{n \cdot h_n^{d_Z}} \sum_{k=1}^n \psi_n^\pi(V_k; Z_\ell, \theta) + \vartheta_n^\pi(Z_\ell, \theta) \right) \cdot \frac{\partial \Xi_S(X_\ell, \theta)}{\partial \Delta}
\end{aligned}$$

From here,

$$\begin{aligned}
& \frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} \varphi_S(V_i, \theta) \cdot \left[\frac{\partial \widehat{\varphi}_S(V_\ell, \theta)}{\partial \Delta} - \frac{\partial \varphi_S(V_\ell, \theta)}{\partial \Delta} \right] \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_\ell \leq U_j\} \\
&= -\frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} \varphi_S(V_i, \theta) \cdot \left(\frac{1}{n \cdot h_n^{d_Z}} \sum_{k=1}^n \psi_n^{\nabla \Delta \pi}(V_k; Z_\ell, \theta) + \vartheta_n^{\nabla \Delta \pi}(Z_\ell, \theta) \right) \cdot \Xi_S(X_\ell, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_\ell \leq U_j\} \\
&\quad - \frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} \varphi_S(V_i, \theta) \cdot \left(\frac{1}{n \cdot h_n^{d_Z}} \sum_{k=1}^n \psi_n^\pi(V_k; Z_\ell, \theta) + \vartheta_n^\pi(Z_\ell, \theta) \right) \cdot \frac{\partial \Xi_S(X_\ell, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_\ell \leq U_j\}
\end{aligned}$$

And,

$$\begin{aligned}
& \frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} \frac{\partial \varphi_S(V_\ell, \theta)}{\partial \Delta} \cdot [\widehat{\varphi}_S(V_i, \theta) - \varphi_S(V_i, \theta)] \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_\ell \leq U_j\} \\
&= -\frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} \frac{\partial \varphi_S(V_\ell, \theta)}{\partial \Delta} \cdot \left(\frac{1}{n \cdot h_n^{d_Z}} \sum_{k=1}^n \psi_n^\pi(V_k; Z_i, \theta) + \vartheta_n^\pi(Z_i, \theta) \right) \cdot \Xi_S(X_i, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_\ell \leq U_j\}
\end{aligned}$$

In what follows, let $(V_1, V_2, V_3, V_4) \sim F_V \otimes F_V \otimes F_V \otimes F_V$ (four randomly drawn observations from our i.i.d sample $(V_i)_{i=1}^n$). Let,

$$\begin{aligned}
g_{\Delta}^a(V_1, V_2, V_3; \theta) &\equiv \varphi_S(V_1, \theta) \cdot \frac{\partial \varphi_S(V_3, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\}, \\
g_{\Delta, n}^b(V_1, V_2, V_3, V_4; \theta) &\equiv \frac{1}{h_n^{d_Z}} \times \left(\varphi_S(V_1, \theta) \cdot \left(\Xi_S(X_3, \theta) \cdot \psi_n^{\nabla \Delta \pi}(V_4; Z_3, \theta) + \frac{\partial \Xi_S(X_3, \theta)}{\partial \Delta} \cdot \psi_n^\pi(V_4; Z_3, \theta) \right) \right. \\
&\quad \left. + \frac{\partial \varphi_S(V_3, \theta)}{\partial \Delta} \cdot \Xi_S(X_1, \theta) \cdot \psi_n^\pi(V_4; Z_1, \theta) \right) \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\},
\end{aligned} \tag{S1.49}$$

$$\begin{aligned}
g_{\Delta, n}^c(V_1, V_2, V_3; \theta) &\equiv \frac{1}{h_n^{d_Z}} \times \left(\varphi_S(V_1, \theta) \cdot \Xi_S(X_3, \theta) \cdot [\psi_n^{\nabla \Delta \pi}(V_1; Z_3, \theta) + \psi_n^{\nabla \Delta \pi}(V_2; Z_3, \theta) + \psi_n^{\nabla \Delta \pi}(V_3; Z_3, \theta)] \right. \\
&\quad \left. + \varphi_S(V_1, \theta) \cdot \frac{\partial \Xi_S(X_3, \theta)}{\partial \Delta} \cdot [\psi_n^\pi(V_1; Z_3, \theta) + \psi_n^\pi(V_2; Z_3, \theta) + \psi_n^\pi(V_3; Z_3, \theta)] \right. \\
&\quad \left. + \frac{\partial \varphi_S(V_3, \theta)}{\partial \Delta} \cdot \Xi_S(X_1, \theta) \cdot [\psi_n^\pi(V_1; Z_1, \theta) + \psi_n^\pi(V_2; Z_1, \theta) + \psi_n^\pi(V_3; Z_1, \theta)] \right) \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\},
\end{aligned} \tag{S1.50}$$

$$\begin{aligned}
g_{\Delta,n}^d(V_1, V_2, V_3; \theta) &\equiv \left(\varphi_S(V_1, \theta) \cdot \left(\Xi_S(X_3, \theta) \cdot \vartheta_n^{\nabla \Delta \pi}(Z_3, \theta) + \frac{\partial \Xi_S(X_3, \theta)}{\partial \Delta} \cdot \vartheta_n^\pi(Z_3, \theta) \right) \right. \\
&\quad \left. + \frac{\partial \varphi_S(V_3, \theta)}{\partial \Delta} \cdot \Xi_S(X_1, \theta) \cdot \vartheta_n^\pi(Z_1, \theta) \right) \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\}, \\
g_{\Delta,n}^e(V_1, V_2, V_3; \theta) &\equiv \left(\Xi_S(X_1, \theta) \cdot \Xi_S(X_3, \theta) \cdot [\widehat{\pi}(Z_1, \theta) - \pi(Z_1, \theta)] \cdot \left[\frac{\partial \widehat{\pi}(Z_3, \theta)}{\partial \Delta} - \frac{\partial \pi(Z_3, \theta)}{\partial \Delta} \right] \right. \\
&\quad \left. + \frac{\partial \Xi_S(X_3, \theta)}{\partial \Delta} \cdot \Xi_S(X_1, \theta) \cdot [\widehat{\pi}(Z_1, \theta) - \pi(Z_1, \theta)] \cdot [\widehat{\pi}(Z_3, \theta) - \pi(Z_3, \theta)] \right) \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\}, \\
&\tag{S1.51} \\
g_{\Delta,n}^f(V_1, V_2; \theta) &\equiv \left(\widehat{\varphi}_S(V_1, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_1, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} + \widehat{\varphi}_S(V_2, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_1, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \right. \\
&\quad \left. + \widehat{\varphi}_S(V_1, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_2, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \right), \\
g_{\Delta,n}^g(V_1; \theta) &\equiv \widehat{\varphi}_S(V_1, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_1, \theta)}{\partial \Delta} \cdot \mathbb{1}\{Z_1 \in \mathcal{Z}\}. \\
&\tag{S1.52}
\end{aligned}$$

Using these expressions, (S1.47) becomes¹,

$$\begin{aligned}
\frac{\partial \widehat{Q}_{S,\mathcal{Z}}(\theta)}{\partial \Delta} &= \frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i,j} g_{\Delta,n}^a(V_i, V_j, V_\ell; \theta) - \frac{1}{n^4} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i,j} \sum_{k \neq i,j,\ell} g_{\Delta,n}^b(V_i, V_j, V_\ell, V_k; \theta) \\
&\quad - \frac{1}{n} \times \left(\frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i,j} g_{\Delta,n}^c(V_i, V_j, V_\ell; \theta) \right) - \frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i,j} g_{\Delta,n}^d(V_i, V_j, V_\ell; \theta) \\
&\quad + \frac{1}{n^3} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i,j} g_{\Delta,n}^e(V_i, V_j, V_\ell; \theta) + \frac{1}{n} \times \left(\frac{1}{n^2} \sum_{j=1}^n \sum_{i \neq j} g_{\Delta,n}^f(V_i, V_j; \theta) \right) + \frac{1}{n^2} \times \left(\frac{1}{n} \sum_{j=1}^n g_{\Delta,n}^g(V_j; \theta) \right) \\
&\tag{S1.53}
\end{aligned}$$

¹Recall that $\mathbb{1}\{U_j \leq u\} \equiv \mathbb{1}\{U_j \in \mathcal{Z}\} \times \mathbb{1}\{U_j \leq u\}$, so $\mathbb{1}_{\mathcal{Z}}\{U_j \leq U_j\} = \mathbb{1}\{U_j \in \mathcal{Z}\}$, which explains the structure of $g_{\Delta,n}^g(V_j, \theta)$.

Let

$$\begin{aligned}
\mathcal{V}_{\Delta,n}^a(\theta) &\equiv \frac{1}{n \cdot (n-1) \cdot (n-2)} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i,j} g_{\Delta,n}^a(V_i, V_j, V_\ell; \theta), \\
\mathcal{V}_{\Delta,n}^b(\theta) &\equiv \frac{1}{n \cdot (n-1) \cdot (n-2) \cdot (n-3)} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i,j} \sum_{k \neq i,j,\ell} g_{\Delta,n}^b(V_i, V_j, V_\ell, V_k; \theta), \\
\mathcal{V}_{\Delta,n}^c(\theta) &\equiv \frac{1}{n \cdot (n-1) \cdot (n-2)} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i,j} g_{\Delta,n}^c(V_i, V_j, V_\ell; \theta), \\
\mathcal{V}_{\Delta,n}^d(\theta) &\equiv \frac{1}{n \cdot (n-1) \cdot (n-2)} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i,j} g_{\Delta,n}^d(V_i, V_j, V_\ell; \theta), \\
\mathcal{V}_{\Delta,n}^e(\theta) &\equiv \frac{1}{n \cdot (n-1) \cdot (n-2)} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i,j} g_{\Delta,n}^e(V_i, V_j, V_\ell; \theta), \\
\mathcal{V}_{\Delta,n}^f(\theta) &\equiv \frac{1}{n \cdot (n-1)} \sum_{j=1}^n \sum_{i \neq j} g_{\Delta,n}^f(V_i, V_j; \theta), \\
\mathcal{V}_{\Delta,n}^g(\theta) &\equiv \frac{1}{n} \sum_{j=1}^n \sum_{i \neq j} g_{\Delta,n}^g(V_j; \theta).
\end{aligned}$$

Then (S1.53) can be re-expressed as,

$$\begin{aligned}
&\frac{\partial \widehat{Q}_{S,\mathcal{Z}}(\theta)}{\partial \Delta} = \\
&\left(\frac{(n-1)(n-2)}{n^2} \right) \cdot \mathcal{V}_{\Delta,n}^a(\theta) - \left(\frac{(n-1)(n-2)(n-3)}{n^3} \right) \cdot \mathcal{V}_{\Delta,n}^b(\theta) - \left(\frac{1}{n} \right) \times \left(\frac{(n-1)(n-2)}{n^2} \right) \cdot \mathcal{V}_{\Delta,n}^c(\theta) \\
&- \left(\frac{(n-1)(n-2)}{n^2} \right) \cdot \mathcal{V}_{\Delta,n}^d(\theta) + \left(\frac{(n-1)(n-2)}{n^2} \right) \cdot \mathcal{V}_{\Delta,n}^e(\theta) + \left(\frac{1}{n} \right) \times \left(\frac{n-1}{n} \right) \cdot \mathcal{V}_{\Delta,n}^f(\theta) + \left(\frac{1}{n^2} \right) \times \mathcal{V}_{\Delta,n}^g(\theta)
\end{aligned} \tag{S1.54}$$

Next, we will analyze each term in (S1.54). Let us begin with $\mathcal{V}_{\Delta,n}^c(\theta)$. By the bounded-variation properties of our kernel function described in Assumption E1 and the Lipschitz-restrictions in Assumption E3, Nolan and Pollard (1987, Lemma 22), Pakes and Pollard (1989, Example 2.10, Lemma 2.13, Lemma 2.14) imply that the class of functions involved is *Euclidean* (see Sherman (1994, Definition 3)), and given the existence of $2+\delta$ -moments in Assumptions I1 and E3, and the boundedness restrictions in part (iii) of Assumption G1, the corresponding envelope for the class also has finite $2+\delta$ -moments. From here, Sherman

(1994, Corollary 4) can be invoked to show that $\sup_{\theta \in \Theta} |\mathcal{V}_{\Delta,n}^c(\theta)| = O_p\left(\frac{1}{h_n^{d_Z}}\right)$. Therefore,

$$\sup_{\theta \in \Theta} \left| \underbrace{\left(\frac{1}{n}\right) \times \left(\frac{(n-1)(n-2)}{n^2}\right)}_{\rightarrow 1} \cdot \mathcal{V}_{\Delta,n}^c(\theta) \right| = O_p\left(\frac{1}{n \cdot h_n^{d_Z}}\right) = o_p\left(n^{-1/2}\right), \quad (\text{S1.55})$$

where the last equality follows from the bandwidth convergence restrictions in Assumption E1. Next, from the asymptotic properties of the remainders $\mathfrak{g}_n^\pi(z, \theta)$ and $\mathfrak{g}_n^{\nabla \Delta \pi}(z, \theta)$ described in (S1.10) and (S1.23), and the integrability and boundedness restrictions in Assumptions G1 and E3, we have²

$$\begin{aligned} & \sup_{\theta \in \Theta} |\mathcal{V}_{\Delta,n}^d(\theta)| \leq \\ & \underbrace{\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\mathfrak{g}_n^{\nabla \Delta \pi}(z, \theta)|}_{=o_p(n^{-1/2})} \times \underbrace{\sup_{\theta \in \Theta} \left(\frac{1}{n \cdot (n-1) \cdot (n-2)} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} |\varphi_S(V_i, \theta)| \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_\ell \leq U_j\} \right)}_{=O_p(1)} \\ & + \underbrace{\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\mathfrak{g}_n^\pi(z, \theta)|}_{=o_p(n^{-1/2})} \times \underbrace{\sup_{\theta \in \Theta} \left(\frac{1}{n \cdot (n-1) \cdot (n-2)} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} |\varphi_S(V_i, \theta)| \cdot \left\| \frac{\partial \Xi_S(X_\ell, \theta)}{\partial \Delta} \right\| \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_\ell \leq U_j\} \right)}_{=O_p(1)} \\ & + \underbrace{\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\mathfrak{g}_n^\pi(z, \theta)|}_{=o_p(n^{-1/2})} \times \underbrace{\sup_{\theta \in \Theta} \left(\frac{1}{n \cdot (n-1) \cdot (n-2)} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} \left\| \frac{\partial \varphi(X_\ell, \theta)}{\partial \Delta} \right\| \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_\ell \leq U_j\} \right)}_{=O_p(1)} \\ & = o_p\left(n^{-1/2}\right). \end{aligned}$$

Thus,

$$\sup_{\theta \in \Theta} |\mathcal{V}_{\Delta,n}^d(\theta)| = o_p\left(n^{-1/2}\right) \implies \sup_{\theta \in \Theta} \left| \underbrace{\left(\frac{(n-1)(n-2)}{n^2}\right)}_{\rightarrow 1} \cdot \mathcal{V}_{\Delta,n}^d(\theta) \right| = o_p\left(n^{-1/2}\right) \quad (\text{S1.56})$$

Combining the uniform rate of convergence results in equations (S1.11) and (S1.24) with the boundedness and existence-of-moments restrictions in Assumptions G1, I1 and E3, we

²Recall that $|\Xi_S(x, \theta)| \leq 1 \ \forall (x, \theta)$.

have

$$\begin{aligned}
& \sup_{\theta \in \Theta} |\mathcal{V}_{\Delta,n}^e(\theta)| \leq \\
& \underbrace{\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\widehat{\pi}(z, \theta) - \pi(z, \theta)|}_{=o_p(n^{-1/4})} \times \underbrace{\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left| \frac{\partial \widehat{\pi}(z, \theta)}{\partial \Delta} - \frac{\partial \pi(z, \theta)}{\partial \Delta} \right|}_{=o_p(n^{-1/4})} \\
& \underbrace{\hspace{10em}}_{=o_p(n^{-1/2})} \\
& + \underbrace{\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\widehat{\pi}(z, \theta) - \pi(z, \theta)|}_{=o_p(n^{-1/4})} \times \underbrace{\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} |\widehat{\pi}(z, \theta) - \pi(z, \theta)|}_{=o_p(n^{-1/4})} \\
& \underbrace{\hspace{10em}}_{=o_p(n^{-1/2})} \\
& \times \underbrace{\sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \left(\frac{1}{n \cdot (n-1) \cdot (n-2)} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i, j} \left\| \frac{\partial \Xi_S(X_\ell, \theta)}{\partial \Delta} \right\| \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_\ell \leq U_j\} \right)}_{=O_p(1)} \\
& = o_p(n^{-1/2}).
\end{aligned}$$

Thus,

$$\sup_{\theta \in \Theta} \underbrace{\left| \left(\frac{(n-1)(n-2)}{n^2} \right) \cdot \mathcal{V}_{\Delta,n}^e(\theta) \right|}_{\rightarrow 1} \quad (\text{S1.57})$$

Next we analyze $\mathcal{V}_{\Delta,n}^f(\theta)$. The results in equations (S1.35)-(S1.42) yield

$$\begin{aligned}
& \sup_{\theta \in \Theta} |\mathcal{V}_{\Delta,n}^f(\theta)| = \sup_{\theta \in \Theta} \left| \frac{1}{n \cdot (n-1)} \sum_{j=1}^n \sum_{i \neq j} g_{\Delta,n}^f(V_i, V_j; \theta) \right| = O_p(1). \\
& \sup_{\theta \in \Theta} |\mathcal{V}_{\Delta,n}^f(\theta)| \\
& = \sup_{\theta \in \Theta} \left| \frac{1}{n \cdot (n-1)} \sum_{j=1}^n \sum_{i \neq j} \left(\widehat{\varphi}_S(V_i, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} + \widehat{\varphi}_S(V_j, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \right. \right. \\
& \quad \left. \left. + \widehat{\varphi}_S(V_i, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_j, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \right) \right| = O_p(1).
\end{aligned}$$

Therefore,

$$\sup_{\theta \in \Theta} \left| \underbrace{\left(\frac{1}{n} \right) \times \left(\frac{n-1}{n} \right)}_{\rightarrow 1} \cdot \mathcal{V}_{\Delta,n}^f(\theta) \right| = O_p\left(\frac{1}{n}\right) = o_p\left(n^{-1/2}\right), \quad (\text{S1.58})$$

For $\mathcal{V}_{\Delta,n}^g(\theta)$, we note once again that the results in equations (S1.35)-(S1.42) yield,

$$\sup_{\theta \in \Theta} |\mathcal{V}_{\Delta,n}^g(\theta)| = \sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{j=1}^n \widehat{\varphi}_S(V_j, \theta) \cdot \frac{\partial \widehat{\varphi}_S(V_j, \theta)}{\partial \Delta} \cdot \mathbb{1}\{Z_j \in \mathcal{Z}\} \right| = O_p(1).$$

Thus,

$$\sup_{\theta \in \Theta} \left| \left(\frac{1}{n^2} \right) \times \mathcal{V}_{\Delta,n}^g(\theta) \right| = O_p\left(\frac{1}{n^2}\right) = o_p\left(n^{-1/2}\right) \quad (\text{S1.59})$$

Next, we will analyze the U-statistics $\mathcal{V}_{\Delta,n}^a(\theta)$ and $\mathcal{V}_{\Delta,n}^b(\theta)$ through their *Hoeffding decompositions* (see Serfling (1980, pages 177-178) or Sherman (1994, equations (6)-(7))). Let us begin with $\mathcal{V}_{\Delta,n}^a(\theta)$. Let

$$\mathcal{T}(\theta) \equiv E \left[g_{\Delta}^a(V_1, V_2, V_3; \theta) \right].$$

Note that $\mathcal{T}(\theta_0) = 0$ since,

$$\begin{aligned} \mathcal{T}(\theta_0) &= E \left[\varphi_S(V_1, \theta_0) \cdot \frac{\partial \varphi_S(V_3, \theta_0)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\} \right] \\ &= E \left[\underbrace{E \left[\varphi_S(V_1, \theta_0) \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \mid U_2 \right]}_{\equiv \tau_{\mathcal{Z}}(U_2, \theta_0) = 0 \text{ a.e } U_2} \cdot \frac{\partial \varphi_S(V_3, \theta_0)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\} \right] \\ &= E \left[0 \cdot \frac{\partial \varphi_S(V_3, \theta_0)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\} \right] \\ &= 0. \end{aligned}$$

Next, define,

$$\widetilde{g}_{\Delta}^a(V_1, V_2, V_3; \theta) \equiv \frac{1}{3!} \sum_p g_{\Delta}^a(V_{m_1}, V_{m_2}, V_{m_3}; \theta), \quad (\text{S1.60})$$

where $\sum_p$ denotes the sum over the $3!$ permutations $\{m_1, m_2, m_3\}$ of $\{1, 2, 3\}$. By construction, $\widetilde{g}_{\Delta}^a(V_1, V_2, V_3; \theta)$ is symmetric in (V_1, V_2, V_3) , and $E[\widetilde{g}_{\Delta}^a(V_1, V_2, V_3; \theta)] = \mathcal{T}(\theta)$. We can express,

$$\mathcal{V}_{\Delta,n}^a(\theta) = \binom{n}{3}^{-1} \sum_c \widetilde{g}_{\Delta}^a(V_i, V_j, V_{\ell}; \theta),$$

where $\sum_c$ denotes the sum over the $\binom{n}{3}$ distinct combinations $\{i, j, \ell\}$ from $\{1, \dots, n\}$. Let

$$\bar{g}_\Delta^a(V_1; \theta) \equiv 3 \cdot \left(E \left[\bar{g}_\Delta^a(V_1, V_2, V_3; \theta) \mid V_1 \right] - \mathcal{T}(\theta) \right)$$

Note that, by construction, $E \left[\bar{g}_\Delta^a(V_1; \theta) \right] = 0 \ \forall \ \theta$. Also note that evaluated at θ_0 , we have,

$$\bar{g}_\Delta^a(V_1; \theta_0) = \varphi_S(V_1, \theta_0) \cdot E \left[E \left[\frac{\partial \varphi_S(V_3, \theta_0)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}\{U_3 \leq U_2\}} \mid U_2 \right] \cdot \mathbb{1}\{U_1 \leq U_2\} \mid U_1 \right]$$

The Hoeffding decomposition of $\mathcal{V}_{\Delta, n}^a(\theta)$ (see Serfling (1980, pages 177-178)) is given by,

$$\mathcal{V}_{\Delta, n}^a(\theta) = \mathcal{T}(\theta) + \frac{1}{n} \sum_{i=1}^n \bar{g}_\Delta^a(V_i; \theta) + \mathcal{U}_n^a(\theta),$$

where $\mathcal{U}_n^a(\theta)$ is a linear combination of *degenerate* U-statistics of orders 2 and 3. By the Lipschitz-restrictions in Assumption E3, Nolan and Pollard (1987, Lemma 22), Pakes and Pollard (1989, Example 2.10, Lemma 2.13, Lemma 2.14) imply that the class of functions involved is *Euclidean* (see Sherman (1994, Definition 3)), and given the existence of $2 + \delta$ -moments in Assumptions I1 and E3, and the boundedness restrictions in part (iii) of Assumption G1, the corresponding envelope for the class also has finite $2 + \delta$ -moments. From here, Sherman (1994, Corollary 4) yields,

$$\sup_{\theta \in \Theta} |\mathcal{U}_n^a(\theta)| = O_p\left(\frac{1}{n}\right) = o_p\left(n^{-1/2}\right).$$

Thus,

$$\mathcal{V}_{\Delta, n}^a(\theta) = \mathcal{T}(\theta) + \frac{1}{n} \sum_{i=1}^n \bar{g}_\Delta^a(V_i; \theta) + \mathcal{U}_n^a(\theta), \quad \text{where} \quad \sup_{\theta \in \Theta} |\mathcal{U}_n^a(\theta)| = o_p\left(n^{-1/2}\right). \quad (\text{S1.61})$$

And evaluated at θ_0 , we have $\mathcal{T}(\theta_0) = 0$, and

$$\mathcal{V}_{\Delta, n}^a(\theta_0) = \frac{1}{n} \sum_{i=1}^n \bar{g}_\Delta^a(V_i; \theta_0) + o_p\left(n^{-1/2}\right), \quad \text{where}$$

$$\bar{g}_\Delta^a(V_1; \theta_0) = \varphi_S(V_1, \theta_0) \cdot E \left[E \left[\frac{\partial \varphi_S(V_3, \theta_0)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}\{U_3 \leq U_2\}} \mid U_2 \right] \cdot \mathbb{1}\{U_1 \leq U_2\} \mid U_1 \right].$$

$\sup_{\theta \in \Theta} |\mathcal{T}(\theta)| < \infty$ under our restrictions. Thus, from (S1.61), $\sup_{\theta \in \Theta} |\mathcal{V}_{\Delta,n}^a(\theta)| = O_p(1)$ and,

$$\sup_{\theta \in \Theta} \underbrace{\left| \left(\frac{2-3n}{n^2} \right) \cdot \mathcal{V}_{\Delta,n}^a(\theta) \right|}_{=O\left(\frac{1}{n}\right)} = O_p\left(\frac{1}{n}\right).$$

From here, it follows that,

$$\left(\frac{(n-1)(n-2)}{n^2} \right) \cdot \mathcal{V}_{\Delta,n}^a(\theta) = \mathcal{V}_{\Delta,n}^a(\theta) + \underbrace{\left(\frac{2-3n}{n^2} \right) \cdot \mathcal{V}_{\Delta,n}^a(\theta)}_{\equiv \mathfrak{S}_n^a(\theta)} = \mathcal{T}(\theta) + \frac{1}{n} \sum_{i=1}^n \bar{g}_{\Delta}^a(V_i; \theta) + \mathfrak{S}_n^a(\theta),$$

$$\text{where } \sup_{\theta \in \Theta} |\mathfrak{S}_n^a(\theta)| = O_p\left(\frac{1}{n}\right) = o_p\left(n^{-1/2}\right).$$

In particular, evaluated at θ_0 ,

$$\begin{aligned} \left(\frac{(n-1)(n-2)}{n^2} \right) \cdot \mathcal{V}_{\Delta,n}^a(\theta_0) &= \frac{1}{n} \sum_{i=1}^n \bar{g}_{\Delta}^a(V_i; \theta_0) + o_p\left(n^{-1/2}\right), \quad \text{where} \\ \bar{g}_{\Delta}^a(V_1; \theta_0) &= \varphi_S(V_1, \theta_0) \cdot E \left[E \left[\frac{\partial \varphi_S(V_3, \theta_0)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}\{U_3 \leq U_2\}} \mid U_2 \right] \cdot \mathbb{1}_{\mathcal{Z}\{U_1 \leq U_2\}} \mid U_1 \right] \end{aligned} \quad (\text{S1.62})$$

Next we analyze the Hoeffding decomposition of $\mathcal{V}_{\Delta,n}^b(\theta)$, the last remaining term to study in (S1.54). Note first that, from its definition in (S1.49), we have

$$E \left[g_{\Delta,n}^b(V_1, V_2, V_3, V_4; \theta) \mid V_1, V_2, V_3 \right] = 0 \quad \forall \theta,$$

and therefore $E \left[g_{\Delta,n}^b(V_1, V_2, V_3, V_4; \theta) \right] = 0 \quad \forall \theta$. This follows because

$$E \left[\psi_n^{\nabla \Delta \pi}(V_4; Z_3, \theta) \mid Z_3 \right] = E \left[\psi_n^{\pi}(V_4; Z_3, \theta) \mid Z_3 \right] = E \left[\psi_n^{\pi}(V_4; Z_1, \theta) \mid Z_1 \right] = 0 \quad \forall \theta.$$

Define,

$$\widetilde{g}_{\Delta,n}^b(V_1, V_2, V_3, V_4; \theta) \equiv \frac{1}{4!} \sum_p g_{\Delta,n}^b(V_{m_1}, V_{m_2}, V_{m_3}, V_{m_4}; \theta),$$

where $\sum_p$ denotes the sum over the $4!$ permutations $\{m_1, m_2, m_3, m_4\}$ of $\{1, 2, 3, 4\}$. By construction, $\widetilde{g}_{\Delta,n}^b(V_1, V_2, V_3, V_4; \theta)$ is symmetric in (V_1, V_2, V_3, V_4) , and $E \left[\widetilde{g}_{\Delta,n}^b(V_1, V_2, V_3, V_4; \theta) \right] = 0 \quad \forall \theta$. We can express,

$$\mathcal{V}_{\Delta,n}^b(\theta) = \binom{n}{4}^{-1} \sum_c \widetilde{g}_{\Delta,n}^b(V_i, V_j, V_\ell, V_k; \theta),$$

where $\sum_c$ denotes the sum over the $\binom{n}{4}$ distinct combinations $\{i, j, \ell, k\}$ from $\{1, \dots, n\}$. Let

$$\bar{g}_{\Delta,n}^b(V_1; \theta) \equiv 4 \cdot E \left[\bar{g}_{\Delta,n}^b(V_1, V_2, V_3, V_4; \theta) \mid V_1 \right].$$

By inspection, we can see that,

$$\begin{aligned} E \left[\bar{g}_{\Delta,n}^b(V_1, V_2, V_3, V_4; \theta) \mid V_1 \right] &= \frac{3!}{4!} E \left[g_{\Delta,n}^b(V_2, V_3, V_4, V_1; \theta) \mid V_1 \right] \\ &= \frac{1}{4} \times \left(E \left[\varphi_S(V_2, \theta) \cdot \Xi_S(X_4, \theta) \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\nabla \Delta \pi}(V_1; Z_4, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \right. \\ &\quad + E \left[\frac{\partial \Xi_S(X_4, \theta)}{\partial \Delta} \cdot \varphi_S(V_2, \theta) \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\pi}(V_1; Z_4, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \\ &\quad \left. + E \left[\frac{\partial \varphi_S(V_4, \theta)}{\partial \Delta} \cdot \Xi_S(X_2, \theta) \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\pi}(V_1; Z_2, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \right) \end{aligned}$$

Therefore,

$$\begin{aligned} \bar{g}_{\Delta,n}^b(V_1; \theta) &= E \left[\varphi_S(V_2, \theta) \cdot \Xi_S(X_4, \theta) \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\nabla \Delta \pi}(V_1; Z_4, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \\ &\quad + E \left[\frac{\partial \Xi_S(X_4, \theta)}{\partial \Delta} \cdot \varphi_S(V_2, \theta) \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\pi}(V_1; Z_4, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \\ &\quad + E \left[\frac{\partial \varphi_S(V_4, \theta)}{\partial \Delta} \cdot \Xi_S(X_2, \theta) \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\pi}(V_1; Z_2, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right]. \end{aligned} \tag{S1.63}$$

Note that, evaluated at θ_0 ,

$$\begin{aligned} &E \left[\varphi_S(V_2, \theta_0) \cdot \Xi_S(X_4, \theta_0) \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\nabla \Delta \pi}(V_1; Z_4, \theta_0) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \\ &= E \left[\underbrace{E[\varphi_S(V_2, \theta_0) \mid U_2]}_{=0} \cdot \Xi_S(X_4, \theta_0) \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\nabla \Delta \pi}(V_1; Z_4, \theta_0) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] = 0, \\ &E \left[\frac{\partial \Xi_S(X_4, \theta_0)}{\partial \Delta} \cdot \varphi_S(V_2, \theta_0) \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\pi}(V_1; Z_4, \theta_0) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \\ &= E \left[\frac{\partial \Xi_S(X_4, \theta_0)}{\partial \Delta} \cdot \underbrace{E[\varphi_S(V_2, \theta_0) \mid U_2]}_{=0} \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\pi}(V_1; Z_4, \theta_0) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] = 0. \end{aligned}$$

Thus, evaluated at θ_0 ,

$$\begin{aligned}\bar{g}_{\Delta,n}^b(V_1; \theta_0) &= E \left[\frac{\partial \varphi_S(V_4, \theta_0)}{\partial \Delta} \cdot \Xi_S(X_2, \theta_0) \cdot \frac{1}{h_n^{d_Z}} \psi_n^\pi(V_1; Z_2, \theta_0) \cdot \mathbb{1}_{\mathcal{Z}\{U_2 \leq U_3\}} \cdot \mathbb{1}_{\mathcal{Z}\{U_4 \leq U_3\}} \middle| V_1 \right] \\ &= E \left[\frac{\partial \varphi_S(V_4, \theta_0)}{\partial \Delta} \cdot E \left[\Xi_S(X_2, \theta_0) \cdot \frac{1}{h_n^{d_Z}} \psi_n^\pi(V_1; Z_2, \theta_0) \cdot \mathbb{1}_{\mathcal{Z}\{U_2 \leq U_3\}} \middle| V_1, U_3 \right] \cdot \mathbb{1}_{\mathcal{Z}\{U_4 \leq U_3\}} \middle| V_1 \right].\end{aligned}\tag{S1.64}$$

The Hoeffding decomposition of $\mathcal{V}_{\Delta,n}^b(\theta)$ is given by,

$$\mathcal{V}_{\Delta,n}^b(\theta) = \frac{1}{n} \sum_{i=1}^n \bar{g}_{\Delta,n}^b(V_i; \theta) + \mathcal{U}_n^b(\theta),$$

where $\mathcal{U}_n^b(\theta)$ is a linear combination of degenerate U-statistics of orders 2, 3 and 4. By the bounded-variation properties of our kernel function described in Assumption E1 and the Lipschitz-restrictions in Assumption E3, Nolan and Pollard (1987, Lemma 22), Pakes and Pollard (1989, Example 2.10, Lemma 2.13, Lemma 2.14) imply that the class of functions involved is *Euclidean* (see Sherman (1994, Definition 3)), and given the existence of $2 + \delta$ -moments in Assumptions I1 and E3, and the boundedness restrictions in part (iii) of Assumption G1, the corresponding envelope for the class also has finite $2 + \delta$ -moments. From here, Sherman (1994, Corollary 4) yields

$$\sup_{\theta \in \Theta} |\mathcal{U}_n^b(\theta)| = O_p \left(\frac{1}{n \cdot h_n^{d_Z}} \right) = o_p(n^{-1/2}),$$

where the last equality follows from our bandwidth convergence restrictions since $n^{1/2} \cdot h_n^{d_Z} \rightarrow \infty$. Thus,

$$\mathcal{V}_{\Delta,n}^b(\theta) = \frac{1}{n} \sum_{i=1}^n \bar{g}_{\Delta,n}^b(V_i; \theta) + \mathcal{U}_n^b(\theta), \quad \text{where} \quad \sup_{\theta \in \Theta} |\mathcal{U}_n^b(\theta)| = o_p(n^{-1/2}),$$

and where $\bar{g}_{\Delta,n}^b(V_1; \theta)$ is as described in (S1.63), and it satisfies $E[\bar{g}_{\Delta,n}^b(V_1; \theta)] = 0 \forall \theta$. Let us focus on $\theta = \theta_0$. Note first that,

$$\begin{aligned}\left(\frac{(n-1)(n-2)(n-3)}{n^3} \right) \cdot \mathcal{V}_{\Delta,n}^b(\theta_0) &= \mathcal{V}_{\Delta,n}^b(\theta_0) + \underbrace{\left(\frac{-6n^2 + 11n - 6}{n^3} \right)}_{=O(\frac{1}{n})} \cdot \underbrace{\mathcal{V}_{\Delta,n}^b(\theta_0)}_{=O_p(1)} \\ &= O_p\left(\frac{1}{n}\right) = o_p(n^{-1/2})\end{aligned}$$

Thus,

$$\left(\frac{(n-1)(n-2)(n-3)}{n^3} \right) \cdot \mathcal{V}_{\Delta,n}^b(\theta_0) = \frac{1}{n} \sum_{i=1}^n \bar{g}_{\Delta,n}^b(V_i; \theta_0) + o_p(n^{-1/2}) \quad (\text{S1.65})$$

where $\bar{g}_{\Delta,n}^b(V_1; \theta_0)$ is as described in (S1.64). We are now ready to put together our results for $\frac{\partial \widehat{Q}_{S,\mathcal{Z}}(\theta_0)}{\partial \Delta}$. As before, let $(V_1, V_2, V_3, V_4) \sim F_V \otimes F_V \otimes F_V \otimes F_V$ (four randomly drawn observations from our i.i.d sample $(V_i)_{i=1}^n$). Combining the results in (S1.55), (S1.56), (S1.57), (S1.58), (S1.59), (S1.62) and (S1.65), and plugging them into equation (S1.54), at $\theta = \theta_0$,

$$\begin{aligned} \frac{\partial \widehat{Q}_{S,\mathcal{Z}}(\theta_0)}{\partial \Delta} &= \frac{1}{n} \sum_{i=1}^n \Gamma_{\Delta,n}(V_i; \theta_0) + o_p(n^{-1/2}), \quad \text{where} \\ \Gamma_{\Delta,n}(V_1; \theta_0) &\equiv \bar{g}_{\Delta}^a(V_1; \theta_0) - \bar{g}_{\Delta,n}^b(V_1; \theta_0) \\ &= \varphi_S(V_1, \theta_0) \cdot E \left[E \left[\frac{\partial \varphi_S(V_3, \theta_0)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}\{U_3 \leq U_2\}} \middle| U_2 \right] \cdot \mathbb{1}_{\mathcal{Z}\{U_1 \leq U_2\}} \middle| U_1 \right] \\ &\quad - E \left[\frac{\partial \varphi_S(V_4, \theta_0)}{\partial \Delta} \cdot E \left[\Xi_S(X_2, \theta_0) \cdot \frac{1}{h_n^{d_Z}} \psi_n^\pi(V_1; Z_2, \theta_0) \cdot \mathbb{1}_{\mathcal{Z}\{U_2 \leq U_3\}} \middle| V_1, U_3 \right] \cdot \mathbb{1}_{\mathcal{Z}\{U_4 \leq U_3\}} \middle| V_1 \right]. \end{aligned} \quad (\text{S1.66})$$

Note, once again, that $E[\Gamma_{\Delta,n}(V_1; \theta_0)] = 0$.

S1.5.2 A linear representation result for $\widehat{\Delta}$ (final step in the proof of Proposition 1)

Let

$$\psi_{\Delta,n}(V_i; \theta_0) \equiv -\frac{\partial^2 Q_{S,\mathcal{Z}}(\theta_0)}{\partial \Delta \partial \Delta'}^{-1} \times \left(\Gamma_{\Delta,n}(V_i; \theta_0) + \frac{\partial^2 Q_{S,\mathcal{Z}}(\theta_0)}{\partial \Delta \partial \gamma'} \cdot \psi_\gamma(V_i; \gamma_0) \right)$$

Note that $E[\psi_{\Delta,n}(V; \theta_0)] = 0$. From equation (S1.46), our result in (S1.66) yields,

$$\widehat{\Delta} - \Delta_0 = \frac{1}{n} \sum_{i=1}^n \psi_{\Delta,n}(V_i; \theta_0) + o_p(n^{-1/2}),$$

and $\sqrt{n} \cdot (\widehat{\Delta} - \Delta_0) \xrightarrow{d} \mathcal{N}(0, \Omega_\Delta)$, where $\Omega_\Delta = \lim_{n \rightarrow \infty} E[\psi_{\Delta,n}(V; \theta_0) \cdot \psi_{\Delta,n}(V; \theta_0)']$. This is the result described in Proposition 1. ■

S1.6 An estimator for Ω_Δ , the asymptotic variance of $\sqrt{n} \cdot (\widehat{\Delta} - \Delta_0)$

Based on our result in Section S1.5.2, our proposal is to estimate Ω_Δ , the asymptotic variance of $\sqrt{n} \cdot (\widehat{\Delta} - \Delta_0)$, as $\widehat{\Omega}_\Delta \equiv \frac{1}{n} \sum_{i=1}^n \widehat{\psi}_{\Delta,n}(V_i; \widehat{\theta}) \cdot \widehat{\psi}_{\Delta,n}(V_i; \widehat{\theta})'$, where $\widehat{\psi}_{\Delta,n}(V_i; \widehat{\theta})$ is an estimator of the influence function $\psi_{\Delta,n}(V_i; \theta)$. We estimate the influence function $\psi_n^\pi(V_i; z, \theta)$

described in equations (S1.6)-(S1.10) as follows. Let,

$$\begin{aligned}
\widehat{R}_a^\pi(z, \theta) &\equiv \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n (S_i - \mu_S^{NC}(X_i, \theta)) \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right), \\
\widehat{R}_b^\pi(z, \theta) &\equiv \frac{1}{n \cdot h_n^{d_Z}} \sum_{i=1}^n \Xi_S(X_i, \theta)^2 \cdot K\left(\frac{Z_i - z}{h_n}\right), \\
\widehat{\psi}_{a,n}^\pi(V_i; z, \theta) &\equiv (S_i - \mu_S^{NC}(X_i, \theta)) \cdot \Xi_S(X_i, \theta) \cdot K\left(\frac{Z_i - z}{h_n}\right) - h_n^{d_Z} \cdot \widehat{R}_a^\pi(z, \theta), \\
\widehat{\psi}_{b,n}^\pi(U_i; z, \theta) &\equiv \Xi_S(X_i, \theta)^2 \cdot K\left(\frac{Z_i - z}{h_n}\right) - h_n^{d_Z} \cdot \widehat{R}_b^\pi(z, \theta), \\
\widehat{\psi}_n^\pi(V_i; z, \theta) &\equiv \frac{1}{\widehat{R}_b^\pi(z, \theta)} \cdot \widehat{\psi}_{a,n}^\pi(V_i; z, \theta) - \frac{\widehat{\pi}(z, \theta)}{\widehat{R}_b^\pi(z, \theta)} \cdot \widehat{\psi}_{b,n}^\pi(U_i; z, \theta)
\end{aligned}$$

Next, we have³

$$\frac{\partial^2 \widehat{Q}_{S,Z}(\theta)}{\partial \theta \partial \theta'} = \frac{1}{n} \sum_{j=1}^n \left(\frac{\partial^2 \widehat{\tau}_Z(U_j, \theta)}{\partial \theta \partial \theta'} \cdot \widehat{\tau}_Z(U_j, \theta) + \frac{\partial \widehat{\tau}_Z(U_j, \theta)}{\partial \theta} \cdot \frac{\partial \widehat{\tau}_Z(U_j, \theta)}{\partial \theta} \right)'.$$

Where,

$$\frac{\partial \widehat{\tau}_Z(u, \theta)}{\partial \theta} = \frac{1}{n} \sum_{i=1}^n \frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \theta} \cdot \mathbb{1}_Z\{U_i \leq u\}, \quad \frac{\partial^2 \widehat{\tau}_Z(u, \theta)}{\partial \theta \partial \theta'} = \frac{1}{n} \sum_{i=1}^n \frac{\partial^2 \widehat{\varphi}_S(V_i, \theta)}{\partial \theta \partial \theta'} \cdot \mathbb{1}_Z\{U_i \leq u\}.$$

And,

$$\begin{aligned}
\widehat{\varphi}_S(V_i, \theta) &\equiv S_i - \mu_S^{NC}(X_i, \theta) - \widehat{\pi}(Z_i, \theta) \cdot \Xi_S(X_i, \theta), \\
\frac{\partial \widehat{\varphi}_S(V_i, \theta)}{\partial \theta} &= - \left(\frac{\partial \mu_S^{NC}(X_i, \theta)}{\partial \theta} + \frac{\partial \widehat{\pi}(Z_i, \theta)}{\partial \theta} \cdot \Xi_S(X_i, \theta) + \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta} \cdot \widehat{\pi}(Z_i, \theta) \right), \\
\frac{\partial^2 \widehat{\varphi}_S(V_i, \theta)}{\partial \theta \partial \theta'} &= - \left(\frac{\partial^2 \mu_S^{NC}(X_i, \theta)}{\partial \theta \partial \theta'} + \frac{\partial \widehat{\pi}(Z_i, \theta)}{\partial \theta} \cdot \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta}' + \frac{\partial^2 \widehat{\pi}(Z_i, \theta)}{\partial \theta \partial \theta'} \cdot \Xi_S(X_i, \theta) \right. \\
&\quad \left. + \frac{\partial \Xi_S(X_i, \theta)}{\partial \theta} \cdot \frac{\partial \widehat{\pi}(Z_i, \theta)}{\partial \theta}' + \frac{\partial^2 \Xi_S(X_i, \theta)}{\partial \theta \partial \theta'} \cdot \widehat{\pi}(Z_i, \theta) \right).
\end{aligned}$$

³Note that $\frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \theta \partial \theta'} = E \left[\frac{\partial \tau_Z(U, \theta_0)}{\partial \theta} \cdot \frac{\partial \tau_Z(U, \theta_0)}{\partial \theta} \right]$ (since $\tau_Z(U, \theta_0) = 0$ a.s), and we can estimate $\frac{\partial^2 Q_{S,Z}(\theta_0)}{\partial \theta \partial \theta'}$ as

$$\frac{\partial^2 \widehat{Q}_{S,Z}(\widehat{\theta})}{\partial \theta \partial \theta'} = \frac{1}{n} \sum_{j=1}^n \frac{\partial \widehat{\tau}_Z(U_j, \widehat{\theta})}{\partial \theta} \cdot \frac{\partial \widehat{\tau}_Z(U_j, \widehat{\theta})}{\partial \theta}'.$$

The Jacobian $\frac{\partial \widehat{\pi}(Z_i, \theta)}{\partial \theta}$ and Hessian $\frac{\partial^2 \widehat{\pi}(Z_i, \theta)}{\partial \theta \partial \theta'}$ of our estimated weights $\widehat{\pi}(Z_i, \theta)$ are as described in equations (S1.12)-(S1.28). For $m \in \mathbb{N}$ let $(m)_k \equiv m \cdot (m-1) \cdots (m-k)$. From the definition in (21), and the definition of $\Gamma_{\Delta, n}(V_1; \theta)$ in (S1.66), we estimate

$$\begin{aligned} \widehat{\Gamma}_{\Delta, n}(V_i; \theta) &= \widehat{\varphi}_S(V_i, \theta) \cdot \frac{1}{(n-1)_1} \sum_{j \neq i} \sum_{k \neq i, j} \frac{\partial \widehat{\varphi}_S(V_k, \theta)}{\partial \Delta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_k \leq U_j\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \\ &\quad - \frac{1}{(n-1)_2} \sum_{j \neq i} \sum_{k \neq i, j} \sum_{\ell \neq i, j, k} \frac{\partial \widehat{\varphi}_S(V_\ell, \theta)}{\partial \Delta} \cdot \Xi_S(X_j, \theta) \cdot \frac{1}{h_n^{d_Z}} \widehat{\psi}_n^\pi(V_i; Z_j, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_j \leq U_k\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_\ell \leq U_k\} \end{aligned}$$

Let $\widehat{\psi}_\gamma(V_i; \widehat{\gamma})$ be the estimated MLE influence function for $\widehat{\gamma}$. Using (22), we estimate $\psi_{\Delta, n}(V_i; \theta_0)$ with,

$$\widehat{\psi}_{\Delta, n}(V_i; \widehat{\theta}) \equiv -\frac{\partial^2 \widehat{Q}_{S, \mathcal{Z}}(\widehat{\theta})}{\partial \Delta \partial \Delta'}^{-1} \times \left(\widehat{\Gamma}_{\Delta, n}(V_i; \widehat{\theta}) + \frac{\partial^2 \widehat{Q}_{S, \mathcal{Z}}(\widehat{\theta})}{\partial \Delta \partial \gamma'} \cdot \widehat{\psi}_\gamma(V_i; \widehat{\gamma}) \right)$$

We estimate Ω_Δ , the asymptotic variance of $\sqrt{n} \cdot (\widehat{\Delta} - \Delta_0)$, as $\widehat{\Omega}_\Delta \equiv \frac{1}{n} \sum_{i=1}^n \widehat{\psi}_{\Delta, n}(V_i; \widehat{\theta}) \cdot \widehat{\psi}_{\Delta, n}(V_i; \widehat{\theta})'$. Using the results in Sections S1.1, S1.2, S1.3, S1.4, we see that under the restrictions of Proposition 1, we have

$$\left\| \frac{1}{n} \sum_{i=1}^n \widehat{\psi}_{\Delta, n}(V_i; \widehat{\theta}) \cdot \widehat{\psi}_{\Delta, n}(V_i; \widehat{\theta})' - \frac{1}{n} \sum_{i=1}^n \psi_{\Delta, n}(V_i; \theta_0) \cdot \psi_{\Delta, n}(V_i; \theta_0)' \right\| \xrightarrow{p} 0,$$

so $\|\widehat{\Omega}_\Delta - \Omega_\Delta\| \xrightarrow{p} 0$.

S2 Asymptotic properties of the estimator $\widehat{\theta}$ proposed in Section 3.2.1

Recall that the estimator $\widehat{\theta}$ is described as follows. For each $y \in \mathcal{Y}$ we defined $D(y) \equiv \mathbb{1}\{Y = y\}$. For $\bar{y} \in \{(1, 0), (0, 1)\}$ and any (z, θ) we defined,

$$\begin{aligned} \widehat{\delta}(z, \theta) &\equiv \left(\sum_{i=1}^n \Xi(\bar{y}|X_i, \theta) \cdot \Xi(\bar{y}|X_i, \theta)' K\left(\frac{Z_i - z}{h_n}\right) \right)^{-1} \sum_{i=1}^n \Xi(\bar{y}|X_i, \theta) \cdot (D_i(\bar{y}) - p_3^{NC}(\bar{y}|X_i, \theta)) K\left(\frac{Z_i - z}{h_n}\right), \\ \widehat{\mathbb{P}}(y|U, \theta) &\equiv p_3^{NC}(y|X, \theta) + \widehat{\delta}(Z, \theta)' \Xi(y|X, \theta), \quad \widehat{\varphi}(y|V, \theta) \equiv D(y) - \widehat{\mathbb{P}}(y|U, \theta). \end{aligned}$$

For a pre-specified range $\mathcal{Z} \subseteq \text{Supp}(Z)$, with $\mathbb{1}_{\mathcal{Z}}\{U \leq u\} \equiv \mathbb{1}\{U \leq u, Z \in \mathcal{Z}\}$, we defined $T_{\mathcal{Z}}(y|u, \theta) \equiv E[\varphi(y|V, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U \leq u\}]$, $Q_{Y, \mathcal{Z}}(y|\theta) \equiv \frac{1}{2} \int T_{\mathcal{Z}}(y|u, \theta)^2 dF_U(u) = \frac{1}{2} E[T_{\mathcal{Z}}(y|U, \theta)^2]$, and $Q_{Y, \mathcal{Z}}(\theta) \equiv \sum_{y \in \mathcal{Y}^*} Q_{Y, \mathcal{Z}}(y|\theta)$. We characterized θ_0 as the minimizer of $Q_{Y, \mathcal{Z}}(\theta)$. Our

sample objective function is,

$$\widehat{Q}_{Y,\mathcal{Z}}(\theta) \equiv \sum_{y \in \mathcal{Y}^*} \widehat{Q}_{Y,\mathcal{Z}}(y|\theta), \quad \text{where}$$

$$\widehat{Q}_{Y,\mathcal{Z}}(y|\theta) \equiv \frac{1}{2} \cdot \frac{1}{n} \sum_{j=1}^n \widehat{T}_{\mathcal{Z}}(y|U_j, \theta)^2 = \frac{1}{2} \cdot \frac{1}{n} \sum_{j=1}^n \left[\frac{1}{n} \sum_{i=1}^n \widehat{\varphi}(y|V_i, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_i \leq U_j\} \right]^2,$$

Our proposed estimator for θ_0 is $\widehat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmin}} \widehat{Q}_{Y,\mathcal{Z}}(\theta)$. This estimator is analogous to the one we studied in Section 2.6 for the pure-strategies case. Accordingly, we will maintain Assumptions G3, I2, I5, E1, E4, and we will properly modify Assumptions E2 and E3 based on the difference in the specifications of equations (11) and (28). Consider the following.

Assumption E2' (A modified version of Assumption E2)

(i) Let M be the integer described in Assumption E1. The following functionals are M -times continuously differentiable with respect to $\mathbf{z}$, with bounded derivatives for all $(\mathbf{z}, \theta) \in \mathcal{Z} \times \Theta$,

$$\cdot f_{\mathcal{Z}}(\mathbf{z})$$

Element-wise:

$$\cdot \mu_I^{\Xi_Y}(y|\mathbf{z}, \theta) \equiv E[\Xi(y|X, \theta)(D(y) - p_3^{NC}(y|X, \theta)) | Z = \mathbf{z}] \quad \forall y \in \mathcal{Y}$$

$$\cdot \mu_{II,1}^{\Xi_Y}(y|\mathbf{z}, \theta) \equiv E[\Xi(y|X, \theta) | Z = \mathbf{z}] \quad \forall y \in \mathcal{Y}$$

$$\cdot \mu_{II,2}^{\Xi_Y}(y|\mathbf{z}, \theta) \equiv E[\Xi(y|X, \theta) \cdot \Xi(y|X, \theta)' | Z = \mathbf{z}] \quad \forall y \in \mathcal{Y}$$

$$\cdot \mu_{III,\theta_\ell}^{\Xi_Y}(y|\mathbf{z}, \theta) \equiv E \left[\frac{\partial \Xi(y|X, \theta)}{\partial \theta_\ell} \cdot (D(y) - p_3^{NC}(y|X, \theta)) \mid Z = \mathbf{z} \right] \quad (\text{for } \ell = 1, \dots, d_\theta \text{ and } \forall y \in \mathcal{Y})$$

$$\cdot \mu_{IV,\theta_\ell}^{\Xi_Y}(y|\mathbf{z}, \theta) \equiv E \left[\Xi_Y(y|X, \theta) \cdot \frac{\partial p_3^{NC}(y|X, \theta)}{\partial \theta_\ell} \mid Z = \mathbf{z} \right] \quad (\text{for } \ell = 1, \dots, d_\theta \text{ and } \forall y \in \mathcal{Y})$$

$$\cdot \mu_{V,\theta_\ell}^{\Xi_Y}(y|\mathbf{z}, \theta) \equiv E \left[\Xi_Y(y|X, \theta) \cdot \frac{\partial \Xi_Y(y|X, \theta)'}{\partial \theta_\ell} \mid Z = \mathbf{z} \right] \quad (\text{for } \ell = 1, \dots, d_\theta \text{ and } \forall y \in \mathcal{Y})$$

$$\cdot \mu_{VI,\theta_\ell}^{\Xi_Y}(y|\mathbf{z}, \theta) \equiv E \left[\frac{\partial \Xi_Y(y|X, \theta)}{\partial \theta_\ell} \mid Z = \mathbf{z} \right] \quad (\text{for } \ell = 1, \dots, d_\theta \text{ and } \forall y \in \mathcal{Y})$$

In addition, there exists a $\overline{C} < \infty$ such that each of the functionals described above is bounded above (in a matrix-norm sense) by $\overline{C}$, for all $(\mathbf{z}, \theta) \in \mathcal{Z} \times \Theta$ and, for $\overline{y} \in \{(0, 1), (1, 0)\}$, we have $\left\| \mu_{II,2}^{\Xi_Y}(\overline{y}|\mathbf{z}, \theta)^{-1} \right\| \leq \overline{C} \quad \forall (\mathbf{z}, \theta) \in \mathcal{Z} \times \Theta$. Finally, $\exists \underline{C} > 0$ such that $f_{\mathcal{Z}}(\mathbf{z}) \geq \underline{C}$ for all $\mathbf{z} \in \mathcal{Z}$.

(ii) The following functionals are element-wise continuously differentiable with respect to $\mathbf{z}$ with bounded first derivative for all $(\mathbf{z}, \theta) \in \mathcal{Z} \times \Theta$, $\forall y \in \mathcal{Y}$,

$$\begin{aligned}
\cdot \Upsilon_{I,\theta_\ell,\theta_j}(y|z,\theta) &\equiv E \left[\frac{\partial^2 \Xi_Y(y|X,\theta)}{\partial \theta_j \partial \theta_\ell} \cdot (D(y) - p_3^{NC}(y|X,\theta)) \mid Z = z \right] \quad (\text{for } j, \ell = 1, \dots, d_\theta) \\
\cdot \Upsilon_{II,\theta_\ell,\theta_j}(y|z,\theta) &\equiv E \left[\frac{\partial \Xi_Y(y|X,\theta)}{\partial \theta_\ell} \cdot \frac{\partial p_3^{NC}(y|X,\theta)}{\partial \theta_j} \mid Z = z \right] \quad (\text{for } j, \ell = 1, \dots, d_\theta) \\
\cdot \Upsilon_{III,\theta_\ell,\theta_j}(y|z,\theta) &\equiv E \left[\Xi_Y(y|X,\theta) \cdot \frac{\partial^2 p_3^{NC}(y|X,\theta)}{\partial \theta_j \partial \theta_\ell} \mid Z = z \right] \quad (\text{for } j, \ell = 1, \dots, d_\theta) \\
\cdot \Upsilon_{IV,\theta_\ell,\theta_j}(y|z,\theta) &\equiv E \left[\frac{\partial^2 \Xi_Y(y|X,\theta)}{\partial \theta_j \partial \theta_\ell} \cdot \Xi_Y(y|X,\theta)' \mid Z = z \right] \quad (\text{for } j, \ell = 1, \dots, d_\theta) \\
\cdot \Upsilon_{V,\theta_\ell,\theta_j}(y|z,\theta) &\equiv E \left[\frac{\partial \Xi_Y(y|X,\theta)}{\partial \theta_\ell} \cdot \frac{\partial \Xi_Y(y|X,\theta)'}{\partial \theta_j} \mid Z = z \right] \quad (\text{for } j, \ell = 1, \dots, d_\theta)
\end{aligned}$$

■

Assumption E3' (A modified version of Assumption E3)

Each of the functions below has the following type of Lipschitz property: $\|g(u, \theta) - g(u, \theta')\| \leq \overline{G}(u) \cdot \|\theta - \theta'\| \forall u \in \text{Supp}(U)$ (unless noted otherwise), and $\forall \theta, \theta' \in \Theta$, where $\overline{G}(\cdot)$ is a nonnegative function that satisfies $E[\overline{G}(U)^{2+\delta}] < \infty$ for some $\delta > 0$.

- The Lipschitz property holds for all $x \in \text{Supp}(X)$ and $y \in \mathcal{Y}$ for the following functions: $p_3^{NC}(y|x, \theta)$, $\Xi_Y(y|x, \theta)$, $\frac{\partial p_3^{NC}(y|x, \theta)}{\partial \theta_\ell}$, $\frac{\partial \Xi_Y(y|x, \theta)}{\partial \theta_\ell}$ (for $\ell = 1, \dots, d_\theta$), with $\overline{G}(X)$ satisfying $E[\overline{G}(X)^{2+\delta}] < \infty$ for some $\delta > 0$.
- The Lipschitz property holds for all $z \in \mathcal{Z}$ and $y \in \mathcal{Y}$ for the following functionals (see our definitions in Assumption E2'): $\mu_I^{\Xi_Y}(y|z, \theta)$, $\mu_{II,\kappa}^{\Xi_Y}(y|z, \theta)$ (for $\kappa = 1, 2$), $\mu_{III,\theta_\ell}^{\Xi_Y}(y|z, \theta)$, $\mu_{IV,\theta_\ell}^{\Xi_Y}(y|z, \theta)$, $\mu_{V,\theta_\ell}^{\Xi_Y}(y|z, \theta)$, $\mu_{VI,\theta_\ell}^{\Xi_Y}(y|z, \theta)$ (for $\ell = 1, \dots, d_\theta$). In each case, $\overline{G}(Z)$ satisfies $E[\overline{G}(Z)^{2+\delta}] < \infty$ for some $\delta > 0$.
- The Lipschitz property holds for all $z \in \mathcal{Z}$ and $y \in \mathcal{Y}$ for the following functionals, and for every $\ell, j = 1, \dots, d_\theta$ (see our definitions in Assumption E2'): $\Upsilon_{I,\theta_\ell,\theta_j}(y|z, \theta)$, $\Upsilon_{II,\theta_\ell,\theta_j}(y|z, \theta)$, $\Upsilon_{III,\theta_\ell,\theta_j}(y|z, \theta)$, $\Upsilon_{IV,\theta_\ell,\theta_j}(y|z, \theta)$, $\Upsilon_{V,\theta_\ell,\theta_j}(y|z, \theta)$. In each case, $\overline{G}(Z)$ satisfies $E[\overline{G}(Z)^{2+\delta}] < \infty$ for some $\delta > 0$.

■

Analogous arguments leading to the linear representation results in (S1.10) and (S1.23)

can be used to show that,

$$\begin{aligned}\widehat{\delta}(z, \theta) &= \delta(z, \theta) + \frac{1}{n \cdot h_n^{d_z}} \sum_{i=1}^n \psi_n^\delta(V_i; z, \theta) + \mathfrak{s}_n^\delta(z, \theta), \\ \frac{\partial \widehat{\delta}(z, \theta)}{\partial \theta} &= \frac{\partial \delta(z, \theta)}{\partial \theta} + \frac{1}{n \cdot h_n^{d_z}} \sum_{i=1}^n \psi_n^{\nabla \theta \delta}(V_i; z, \theta) + \mathfrak{s}_n^{\nabla \theta \delta}(z, \theta), \\ \text{where } \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \|\mathfrak{s}_n^\delta(z, \theta)\| &= o_p(n^{-1/2}), \quad \sup_{\substack{\theta \in \Theta \\ z \in \mathcal{Z}}} \|\mathfrak{s}_n^{\nabla \theta \delta}(z, \theta)\| = o_p(n^{-1/2}),\end{aligned}$$

and where $E[\psi_n^\delta(V; z, \theta)] = 0$, $E[\psi_n^{\nabla \theta \delta}(V; z, \theta)] = 0 \forall (z, \theta) \in \mathcal{Z} \times \Theta$. From here, similar steps to those in Sections S1.3-S1.4 in this Supplement can be used to show that, under our restrictions, we have $\sup_{\theta \in \Theta} |\widehat{Q}_{Y, \mathcal{Z}}(\theta) - Q_{Y, \mathcal{Z}}(\theta)| = o_p(1)$, $\sup_{\theta \in \Theta} \left| \frac{\partial^2 \widehat{Q}_{Y, \mathcal{Z}}(\theta)}{\partial \theta \partial \theta'} - \frac{\partial^2 Q_{Y, \mathcal{Z}}(\theta)}{\partial \theta \partial \theta'} \right| = o_p(1)$, and $\widehat{\theta} \xrightarrow{p} \theta_0$. Since $\theta_0 \in \text{int}(\Theta)$, w.p.a.1, $\widehat{\theta}$ satisfies,

$$\widehat{\theta} - \theta_0 = - \left(\frac{\partial^2 Q_{Y, \mathcal{Z}}(\theta_0)}{\partial \theta \partial \theta'} \right)^{-1} + o_p(1) \times \frac{\partial \widehat{Q}_{Y, \mathcal{Z}}(\theta_0)}{\partial \theta} \quad (\text{S2.1})$$

This is analogous to (S1.45). The asymptotic distribution of $\widehat{\theta}$ is obtained from here by proving that $\frac{\partial \widehat{Q}_{Y, \mathcal{Z}}(\theta_0)}{\partial \theta}$ satisfies a linear representation result analogous to the one obtained in Section S1.5.1. Since $\frac{\partial \widehat{Q}_{Y, \mathcal{Z}}(\theta_0)}{\partial \theta} = \sum_{y \in \mathcal{Y}^*} \frac{\partial \widehat{Q}_{Y, \mathcal{Z}}(y|\theta_0)}{\partial \theta}$, we will focus on proving that $\frac{\partial \widehat{Q}_{Y, \mathcal{Z}}(y|\theta_0)}{\partial \theta}$ satisfies a linear representation for each $y \in \mathcal{Y}$. In what follows, let $(V_1, V_2, V_3, V_4) \sim F_V \otimes F_V \otimes F_V \otimes F_V$ (four randomly drawn observations from our i.i.d sample $(V_i)_{i=1}^n$). Let,

$$\begin{aligned}g_\theta^a(y|V_1, V_2, V_3; \theta) &\equiv \frac{\partial \varphi(y|V_3, \theta)}{\partial \theta} \cdot \varphi(y|V_1, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\}, \\ g_{\theta, n}^b(y|V_1, V_2, V_3, V_4; \theta) &\equiv \frac{1}{h_n^{d_z}} \times \left(\left(\psi_n^{\nabla \theta \delta}(V_4; Z_3, \theta)' \Xi(y|X_3, \theta) + \frac{\partial \Xi(y|X_3, \theta)'}{\partial \theta} \psi_n^\delta(V_4; Z_3, \theta) \right) \cdot \varphi(y|V_1, \theta) \right. \\ &\quad \left. + \frac{\partial \varphi(y|V_3, \theta)}{\partial \theta} \cdot \psi_n^\delta(V_4; Z_1, \theta)' \Xi(y|X_1, \theta) \right) \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\}\end{aligned}$$

Since $E[\psi_n^\delta(V; z, \theta)] = 0$, $E[\psi_n^{\nabla \theta \delta}(V; z, \theta)] = 0 \forall (z, \theta) \in \mathcal{Z} \times \Theta$, we have

$$E[g_{\theta, n}^b(y|V_1, V_2, V_3, V_4; \theta)|V_1, V_2, V_3] = 0 \forall \theta \in \Theta, y \in \mathcal{Y}$$

Let

$$\begin{aligned}\mathcal{V}_{\theta,n}^a(y|\theta) &\equiv \frac{1}{n \cdot (n-1) \cdot (n-2)} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i,j} g_{\theta}^a(y|V_i, V_j, V_{\ell}; \theta), \\ \mathcal{V}_{\theta,n}^b(y|\theta) &\equiv \frac{1}{n \cdot (n-1) \cdot (n-2) \cdot (n-3)} \sum_{j=1}^n \sum_{i \neq j} \sum_{\ell \neq i,j} \sum_{k \neq i,j,\ell} g_{\theta,n}^b(y|V_i, V_j, V_{\ell}, V_k; \theta)\end{aligned}$$

Parallel arguments to those leading to the results in equations (S1.54)-(S1.59) now yield,

$$\begin{aligned}\frac{\partial \widehat{Q}_{Y,Z}(y|\theta)}{\partial \Delta} &= \left(\frac{(n-1)(n-2)}{n^2} \right) \cdot \mathcal{V}_{\theta,n}^a(y|\theta) - \left(\frac{(n-1)(n-2)(n-3)}{n^3} \right) \cdot \mathcal{V}_{\theta,n}^b(y|\theta) + \xi_{I,n}^{\nabla \theta Q}(y|\theta), \\ \text{where } \sup_{\theta \in \Theta} \left\| \xi_{I,n}^{\nabla \theta Q}(y|\theta) \right\| &= o_p(n^{-1/2})\end{aligned}\tag{S2.2}$$

From here, the next step is to analyze the U-statistics $\mathcal{V}_{\theta,n}^a(y|\theta)$ and $\mathcal{V}_{\theta,n}^b(y|\theta)$ through their *Hoeffding decompositions* (see Serfling (1980, pages 177-178) or Sherman (1994, equations (6)-(7))). Let us begin with $\mathcal{V}_{\theta,n}^a(y|\theta)$. Let

$$T(y|\theta) \equiv E \left[g_{\theta}^a(y|V_1, V_2, V_3; \theta) \right].$$

Note that $T(y|\theta_0) = 0$ since,

$$\begin{aligned}T(y|\theta_0) &= E \left[\frac{\partial \varphi(y|V_3, \theta_0)}{\partial \theta} \cdot \varphi(y|V_1, \theta_0) \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\} \right] \\ &= E \left[\frac{\partial \varphi(y|V_3, \theta_0)}{\partial \theta} \cdot \underbrace{E \left[\varphi(y|V_1, \theta_0) \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \mid U_2 \right]}_{\equiv T_{\mathcal{Z}}(y|U_2, \theta_0)=0 \text{ a.e } U_2} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\} \right] \\ &= E \left[0 \cdot \frac{\partial \varphi(y|V_3, \theta_0)}{\partial \theta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\} \right] \\ &= 0.\end{aligned}$$

Next, define,

$$\widetilde{g}_{\theta}^a(y|V_1, V_2, V_3; \theta) \equiv \frac{1}{3!} \sum_p g_{\theta}^a(y|V_{m_1}, V_{m_2}, V_{m_3}; \theta),\tag{S2.3}$$

where $\sum_p$ denotes the sum over the $3!$ permutations $\{m_1, m_2, m_3\}$ of $\{1, 2, 3\}$. By construction, $\widetilde{g}_{\theta}^a(y|V_1, V_2, V_3; \theta)$ is symmetric in (V_1, V_2, V_3) , and $E \left[\widetilde{g}_{\theta}^a(y|V_1, V_2, V_3; \theta) \right] = T(y|\theta)$. We can express,

$$\mathcal{V}_{\theta,n}^a(y|\theta) = \binom{n}{3}^{-1} \sum_c \widetilde{g}_{\theta}^a(y|V_i, V_j, V_{\ell}; \theta),$$

where $\sum_c$ denotes the sum over the $\binom{n}{3}$ distinct combinations $\{i, j, \ell\}$ from $\{1, \dots, n\}$. Let

$$\bar{g}_\theta^a(y|V_1; \theta) \equiv 3 \cdot \left(E \left[\bar{g}_\theta^a(y|V_1, V_2, V_3; \theta) \mid V_1 \right] - \mathcal{T}(y|\theta) \right)$$

Note that, by construction, $E[\bar{g}_\theta^a(y|V_1; \theta)] = 0 \forall \theta$. Also note that evaluated at θ_0 , we have,

$$\bar{g}_\theta^a(y|V_1; \theta_0) = E \left[E \left[\frac{\partial \varphi(y|V_3, \theta_0)}{\partial \theta} \cdot \mathbb{1}_{\mathcal{Z}\{U_3 \leq U_2\}} \mid U_2 \right] \cdot \mathbb{1}_{\mathcal{Z}\{U_1 \leq U_2\}} \mid U_1 \right] \cdot \varphi(y|V_1, \theta_0)$$

The Hoeffding decomposition of $\mathcal{V}_{\theta,n}^a(y|\theta)$ (see Serfling (1980, pages 177-178)) is given by,

$$\mathcal{V}_{\theta,n}^a(y|\theta) = \mathcal{T}(y|\theta) + \frac{1}{n} \sum_{i=1}^n \bar{g}_\theta^a(y|V_i; \theta) + \mathcal{U}_n^a(y|\theta),$$

where $\mathcal{U}_n^a(y|\theta)$ is a linear combination of *degenerate* U-statistics of orders 2 and 3. Given our smoothness and integrability restrictions, Nolan and Pollard (1987, Lemma 22), Pakes and Pollard (1989, Example 2.10, Lemma 2.13, Lemma 2.14) imply that the class of functions involved is *Euclidean* (see Sherman (1994, Definition 3)). From here, Sherman (1994, Corollary 4) yields,

$$\sup_{\theta \in \Theta} |\mathcal{U}_n^a(y|\theta)| = O_p\left(\frac{1}{n}\right) = o_p\left(n^{-1/2}\right).$$

Thus,

$$\mathcal{V}_{\theta,n}^a(y|\theta) = \mathcal{T}(y|\theta) + \frac{1}{n} \sum_{i=1}^n \bar{g}_\theta^a(y|V_i; \theta) + \mathcal{U}_n^a(y|\theta), \quad \text{where} \quad \sup_{\theta \in \Theta} |\mathcal{U}_n^a(y|\theta)| = o_p\left(n^{-1/2}\right). \quad (\text{S2.4})$$

And evaluated at θ_0 , we have $\mathcal{T}(\theta_0) = 0$, and

$$\begin{aligned} \mathcal{V}_{\theta,n}^a(y|\theta_0) &= \frac{1}{n} \sum_{i=1}^n \bar{g}_{\theta_0}^a(y|V_i; \theta_0) + o_p\left(n^{-1/2}\right), \quad \text{where} \\ \bar{g}_{\theta_0}^a(y|V_1; \theta_0) &= E \left[E \left[\frac{\partial \varphi(y|V_3, \theta_0)}{\partial \theta} \cdot \mathbb{1}_{\mathcal{Z}\{U_3 \leq U_2\}} \mid U_2 \right] \cdot \mathbb{1}_{\mathcal{Z}\{U_1 \leq U_2\}} \mid U_1 \right] \cdot \varphi(y|V_1, \theta_0) \end{aligned}$$

Under our restrictions, $\sup_{\theta \in \Theta} |\mathcal{T}(y|\theta)| < \infty$. Thus, from (S2.4), we have $\sup_{\theta \in \Theta} |\mathcal{V}_{\theta,n}^a(y|\theta)| = O_p(1)$

and,

$$\sup_{\theta \in \Theta} \underbrace{\left| \left(\frac{2-3n}{n^2} \right) \cdot \mathcal{V}_{\theta,n}^a(y|\theta) \right|}_{=O\left(\frac{1}{n}\right)} = O_p\left(\frac{1}{n}\right).$$

From here, it follows that,

$$\left(\frac{(n-1)(n-2)}{n^2}\right) \cdot \mathcal{V}_{\theta,n}^a(y|\theta) = \mathcal{V}_{\theta,n}^a(y|\theta) + \underbrace{\left(\frac{2-3n}{n^2}\right) \cdot \mathcal{V}_{\theta,n}^a(y|\theta)}_{\equiv \mathfrak{S}_n^a(y|\theta)} = \mathcal{T}(y|\theta) + \frac{1}{n} \sum_{i=1}^n \bar{g}_{\theta}^a(y|V_i; \theta) + \mathfrak{S}_n^a(y|\theta),$$

where $\sup_{\theta \in \Theta} |\mathfrak{S}_n^a(y|\theta)| = O_p\left(\frac{1}{n}\right) = o_p\left(n^{-1/2}\right)$.

In particular, evaluated at θ_0 ,

$$\begin{aligned} \left(\frac{(n-1)(n-2)}{n^2}\right) \cdot \mathcal{V}_{\theta,n}^a(y|\theta_0) &= \frac{1}{n} \sum_{i=1}^n \bar{g}_{\theta}^a(y|V_i; \theta_0) + o_p\left(n^{-1/2}\right), \quad \text{where} \\ \bar{g}_{\theta}^a(y|V_1; \theta_0) &= E \left[E \left[\frac{\partial \varphi(y|V_3, \theta_0)}{\partial \theta} \cdot \mathbb{1}_{\mathcal{Z}\{U_3 \leq U_2\}} \mid U_2 \right] \cdot \mathbb{1}_{\mathcal{Z}\{U_1 \leq U_2\}} \mid U_1 \right] \cdot \varphi(y|V_1, \theta_0) \end{aligned} \quad (\text{S2.5})$$

Next we analyze the Hoeffding decomposition of $\mathcal{V}_{\theta,n}^b(y|\theta)$, the last remaining term to study in (S2.2). As we pointed out previously,

$$E[\mathcal{g}_{\theta,n}^b(y|V_1, V_2, V_3, V_4; \theta) | V_1, V_2, V_3] = 0 \quad \forall \theta \in \Theta, y \in \mathcal{Y}$$

and therefore $E[\mathcal{g}_{\theta,n}^b(y|V_1, V_2, V_3, V_4; \theta)] = 0 \quad \forall \theta \in \Theta, y \in \mathcal{Y}$. Define,

$$\widetilde{\mathcal{g}}_{\theta,n}^b(y|V_1, V_2, V_3, V_4; \theta) \equiv \frac{1}{4!} \sum_p \mathcal{g}_{\theta,n}^b(V_{m_1}, V_{m_2}, V_{m_3}, V_{m_4}; \theta),$$

where $\sum_p$ denotes the sum over the $4!$ permutations $\{m_1, m_2, m_3, m_4\}$ of $\{1, 2, 3, 4\}$. By construction, $\widetilde{\mathcal{g}}_{\theta,n}^b(y|V_1, V_2, V_3, V_4; \theta)$ is symmetric in (V_1, V_2, V_3, V_4) , and $E[\widetilde{\mathcal{g}}_{\theta,n}^b(y|V_1, V_2, V_3, V_4; \theta)] = 0 \quad \forall \theta$. We can express,

$$\mathcal{V}_{\theta,n}^b(y|\theta) = \binom{n}{4}^{-1} \sum_c \widetilde{\mathcal{g}}_{\theta,n}^b(y|V_i, V_j, V_\ell, V_k; \theta),$$

where $\sum_c$ denotes the sum over the $\binom{n}{4}$ distinct combinations $\{i, j, \ell, k\}$ from $\{1, \dots, n\}$. Let

$$\bar{\mathcal{g}}_{\theta,n}^b(y|V_1; \theta) \equiv 4 \cdot E[\widetilde{\mathcal{g}}_{\theta,n}^b(y|V_1, V_2, V_3, V_4; \theta) | V_1].$$

By inspection, we can see that,

$$\begin{aligned}
E \left[\tilde{g}_{\theta,n}^b(y|V_1, V_2, V_3, V_4; \theta) \mid V_1 \right] &= \frac{3!}{4!} E \left[g_{\theta,n}^b(y|V_2, V_3, V_4, V_1; \theta) \mid V_1 \right] \\
&= \frac{1}{4} \times \left(E \left[\frac{1}{h_n^{d_Z}} \psi_n^{\nabla \delta}(V_1; Z_4, \theta)' \Xi(y|X_4, \theta) \cdot \varphi(y|V_2, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \right. \\
&\quad + E \left[\frac{\partial \Xi(y|X_4, \theta)'}{\partial \theta} \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\delta}(V_1; Z_4, \theta) \cdot \varphi(y|V_2, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \\
&\quad \left. + E \left[\frac{\partial \varphi(y|V_4, \theta)}{\partial \theta} \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\delta}(V_1; Z_2, \theta)' \Xi(y|X_2, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \right)
\end{aligned}$$

Therefore,

$$\begin{aligned}
\bar{g}_{\theta,n}^b(y|V_1; \theta) &= E \left[\frac{1}{h_n^{d_Z}} \psi_n^{\nabla \delta}(V_1; Z_4, \theta)' \Xi(y|X_4, \theta) \cdot \varphi(y|V_2, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \\
&\quad + E \left[\frac{\partial \Xi(y|X_4, \theta)'}{\partial \theta} \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\delta}(V_1; Z_4, \theta) \cdot \varphi(y|V_2, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \\
&\quad + E \left[\frac{\partial \varphi(y|V_4, \theta)}{\partial \theta} \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\delta}(V_1; Z_2, \theta)' \Xi(y|X_2, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right].
\end{aligned}$$

Note that, evaluated at θ_0 ,

$$\begin{aligned}
&E \left[\frac{1}{h_n^{d_Z}} \psi_n^{\nabla \delta}(V_1; Z_4, \theta_0)' \Xi(y|X_4, \theta_0) \cdot \varphi(y|V_2, \theta_0) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \\
&= E \left[\frac{1}{h_n^{d_Z}} \psi_n^{\nabla \delta}(V_1; Z_4, \theta_0)' \Xi(y|X_4, \theta_0) \cdot \underbrace{E[\varphi(y|V_2, \theta_0) | U_2]}_{=0} \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] = 0, \\
&E \left[\frac{\partial \Xi(y|X_4, \theta_0)'}{\partial \theta} \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\delta}(V_1; Z_4, \theta_0) \cdot \varphi(y|V_2, \theta_0) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \\
&= E \left[\frac{\partial \Xi(y|X_4, \theta_0)'}{\partial \theta} \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\delta}(V_1; Z_4, \theta_0) \cdot \underbrace{E[\varphi(y|V_2, \theta_0) | U_2]}_{=0} \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] = 0
\end{aligned}$$

Thus, evaluated at θ_0 ,

$$\begin{aligned}
\bar{g}_{\theta,n}^b(y|V_1; \theta_0) &= E \left[\frac{\partial \varphi(y|V_4, \theta)}{\partial \theta} \cdot \frac{1}{h_n^{d_Z}} \psi_n^{\delta}(V_1; Z_2, \theta)' \Xi(y|X_2, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right] \\
&= E \left[\frac{\partial \varphi(y|V_4, \theta)}{\partial \theta} \cdot E \left[\frac{1}{h_n^{d_Z}} \psi_n^{\delta}(V_1; Z_2, \theta)' \Xi(y|X_2, \theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \mid V_1, U_3 \right] \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \mid V_1 \right].
\end{aligned} \tag{S2.6}$$

The Hoeffding decomposition of $\mathcal{V}_{\theta,n}^b(y|\theta)$ is given by,

$$\mathcal{V}_{\theta,n}^b(y|\theta) = \frac{1}{n} \sum_{i=1}^n \bar{g}_{\theta,n}^b(y|V_i; \theta) + \mathcal{U}_n^b(y|\theta),$$

where $\mathcal{U}_n^b(y|\theta)$ is a linear combination of degenerate U-statistics of orders 2, 3 and 4. By the bounded-variation properties of our kernel function described in Assumption E1 and our smoothness and integrability conditions, Nolan and Pollard (1987, Lemma 22), Pakes and Pollard (1989, Example 2.10, Lemma 2.13, Lemma 2.14) imply that the class of functions involved is *Euclidean* (see Sherman (1994, Definition 3)). From here, Sherman (1994, Corollary 4) yields

$$\sup_{\theta \in \Theta} |\mathcal{U}_n^b(y|\theta)| = O_p\left(\frac{1}{n \cdot h_n^{d_Z}}\right) = o_p(n^{-1/2}),$$

where the last equality follows from the restriction $n^{1/2} \cdot h_n^{d_Z} \rightarrow \infty$. Thus,

$$\mathcal{V}_{\theta,n}^b(y|\theta) = \frac{1}{n} \sum_{i=1}^n \bar{g}_{\theta,n}^b(y|V_i; \theta) + \mathcal{U}_n^b(y|\theta), \quad \text{where} \quad \sup_{\theta \in \Theta} |\mathcal{U}_n^b(y|\theta)| = o_p(n^{-1/2}),$$

Let us focus on $\theta = \theta_0$. Note first that,

$$\begin{aligned} \left(\frac{(n-1)(n-2)(n-3)}{n^3}\right) \cdot \mathcal{V}_{\theta,n}^b(y|\theta_0) &= \mathcal{V}_{\theta,n}^b(y|\theta_0) + \underbrace{\left(\frac{-6n^2 + 11n - 6}{n^3}\right)}_{=O(\frac{1}{n})} \cdot \underbrace{\mathcal{V}_{\theta,n}^b(y|\theta_0)}_{=O_p(1)} \\ &\underbrace{\hspace{10em}}_{=O_p(\frac{1}{n})=o_p(n^{-1/2})} \end{aligned}$$

Thus,

$$\left(\frac{(n-1)(n-2)(n-3)}{n^3}\right) \cdot \mathcal{V}_{\theta,n}^b(y|\theta_0) = \frac{1}{n} \sum_{i=1}^n \bar{g}_{\theta,n}^b(y|V_i; \theta_0) + o_p(n^{-1/2}) \quad (\text{S2.7})$$

where $\bar{g}_{\theta,n}^b(y|V_1; \theta_0)$ is as described in (S2.6). Combining our results in (S2.2), (S2.5) and

(S2.7),

$$\begin{aligned}\frac{\partial \widehat{Q}_{Y,\mathcal{Z}}(y|\theta_0)}{\partial \theta} &= \frac{1}{n} \sum_{i=1}^n \Gamma_{\theta,n}(y|V_i;\theta_0) + o_p(n^{-1/2}), \quad \text{where} \\ \Gamma_{\theta,n}(y|V_1;\theta_0) &\equiv \bar{g}_{\theta}^a(y|V_1;\theta_0) - \bar{g}_{\theta,n}^b(y|V_1;\theta_0) \\ &= E \left[E \left[\frac{\partial \varphi(y|V_3,\theta_0)}{\partial \theta} \cdot \mathbb{1}_{\mathcal{Z}}\{U_3 \leq U_2\} \middle| U_2 \right] \cdot \mathbb{1}_{\mathcal{Z}}\{U_1 \leq U_2\} \middle| U_1 \right] \cdot \varphi(y|V_1,\theta_0) \\ &\quad - E \left[\frac{\partial \varphi(y|V_4,\theta)}{\partial \theta} \cdot E \left[\frac{1}{h_n^{d_Z}} \psi_n^\delta(V_1;Z_2,\theta)' \Xi(y|X_2,\theta) \cdot \mathbb{1}_{\mathcal{Z}}\{U_2 \leq U_3\} \middle| V_1, U_3 \right] \cdot \mathbb{1}_{\mathcal{Z}}\{U_4 \leq U_3\} \middle| V_1 \right].\end{aligned}$$

Note, once again, that $E[\Gamma_{\theta,n}(y|V_1;\theta_0)] = 0$. From here,

$$\frac{\partial \widehat{Q}_{Y,\mathcal{Z}}(\theta_0)}{\partial \theta} = \frac{1}{n} \sum_{i=1}^n \Gamma_{\theta,n}(V_i;\theta_0) + o_p(n^{-1/2}), \quad \text{where} \quad \Gamma_{\theta,n}(V_i;\theta_0) \equiv \sum_{y \in \mathcal{Y}^*} \Gamma_{\theta,n}(y|V_i;\theta_0). \quad (\text{S2.8})$$

Thus, $E[\Gamma_{\theta,n}(V_1;\theta_0)] = 0$. Let

$$\psi_{\theta,n}(V_i;\theta_0) \equiv -\frac{\partial^2 Q_{Y,\mathcal{Z}}(\theta_0)}{\partial \theta \partial \theta'}^{-1} \times \Gamma_{\theta,n}(V_i;\theta_0).$$

Note that $E[\psi_{\theta,n}(V;\theta_0)] = 0$. From equation (S2.1), our result in (S2.8) yields,

$$\widehat{\theta} - \theta_0 = \frac{1}{n} \sum_{i=1}^n \psi_{\theta,n}(V_i;\theta_0) + o_p(n^{-1/2}).$$

This is the expression in equation (31) of the paper. ■

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